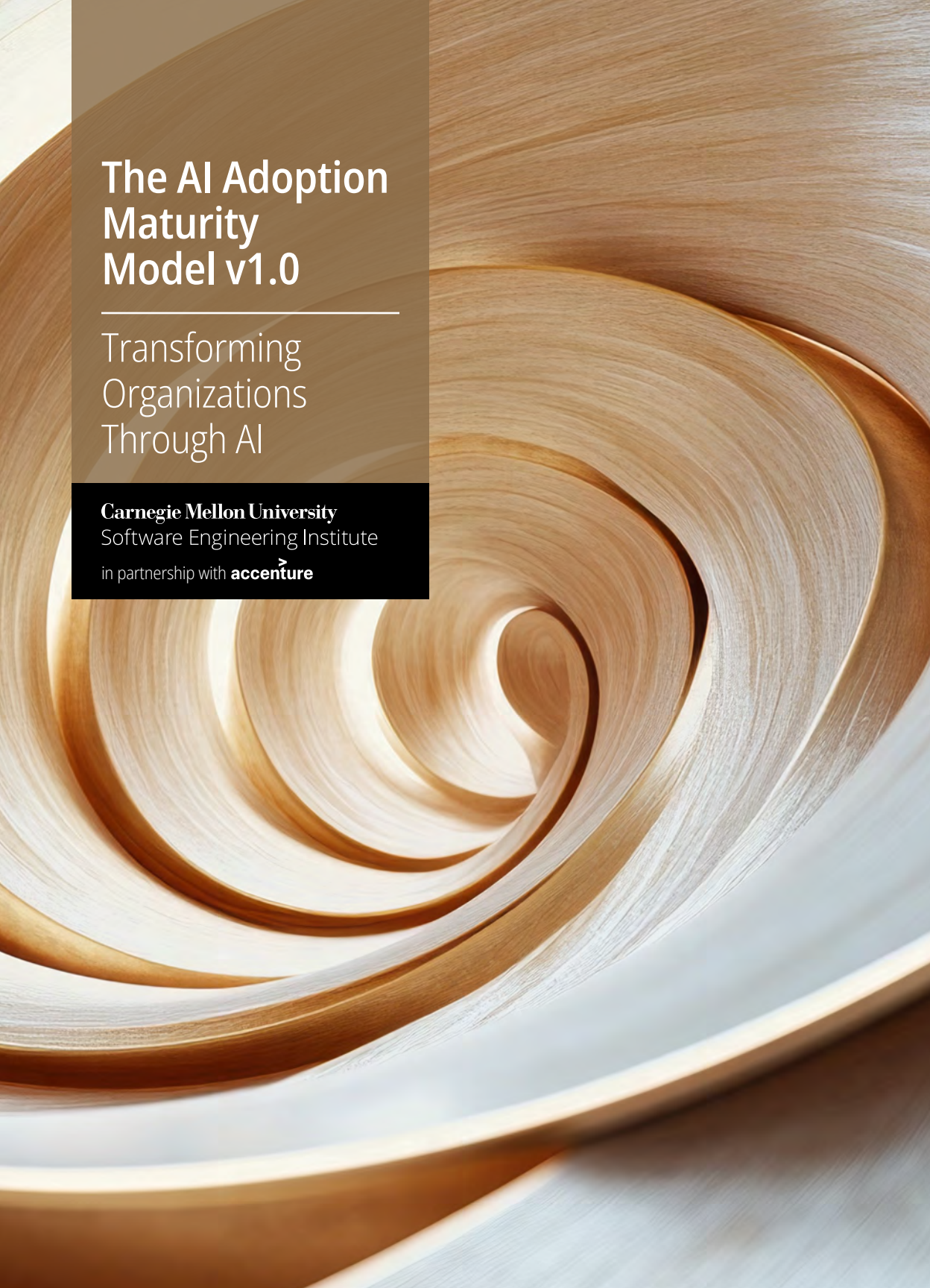




# The AI Adoption Maturity Model v1.0

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Transforming  
Organizations  
Through AI

**Carnegie Mellon University**  
Software Engineering Institute  
in partnership with **accenture**





Organizations achieve lasting AI value and return on investment (ROI) through discipline—not just speed. True AI maturity is not measured by how much AI an organization deploys, but by its ability to build trustworthy and resilient capabilities, rigorous engineering practices, and governance approaches aligned with business outcomes and evolving technological realities. Without maturity, AI adoption risks becoming experimentation at enterprise scale. As a federally funded research and development center (FFRDC), the Carnegie Mellon® University (CMU) Software Engineering Institute (SEI), in collaboration with Accenture, developed the AI Adoption Maturity Model through continuous refinement, real-world application, and community engagement, to help organizations drive sustainable AI transformation and advance the state of the practice.



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Developing an AI adoption maturity model amid one of the fastest technological transformations in history presents two challenges: providing structure in a rapidly evolving landscape, and balancing guidance with the reality that maturity is not a compliance exercise. We developed this maturity model around enduring organizational capabilities rather than transient technologies, viewing maturity as a strategic choice rather than a prescribed destination. The result reflects the collective insights of researchers, practitioners, industry and government leaders, and organizations committed to advancing sustainable, value-driven AI adoption.

The development of the AI Adoption Maturity Model would not have been possible without the dedicated work of many individuals and organizations. First and foremost, we thank Rajendra Prasad, Group Chief Executive of Accenture Technology, whose vision and leadership inspired and shaped this initiative, and Anita Carleton, Director of the Software Solutions Division at the Carnegie Mellon® University (CMU) Software Engineering Institute (SEI), whose deep expertise in maturity models, measurement, and analysis frameworks helped bridge proven approaches with the emerging realities of AI adoption. Majd Sakr, Chief Learning and Research Officer, Accenture and Professor, School of Computer Science, CMU was instrumental in bringing the two organizations together. The collaborative spirit and shared vision established by our Accenture colleagues Satyajit Doctor, Sarah Dugan, Gayathri Pallail, Kaveh Safavi, and Allison Sciotto enabled the tempo we worked with and the model's real-world relevance. Francis Hintermann, Global Managing Director of Accenture Research, and Emily Thornton, Editor-in-Chief of Accenture Research, were invaluable partners in helping gather rigorous, relevant data that captured the broad spectrum of opportunities and challenges that organizations face in their AI adoption journeys.

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Finally, to the many individuals, teams, and stakeholders not named here, thank you. Your feedback, suggestions, support, and countless behind-the-scenes contributions were essential to bringing this work to fruition in just twelve months.

It is only through a collective effort between industry, government, and academia that frameworks like this AI Adoption Maturity Model can be shaped to serve the broader community. While this first phase established the foundation, next phase will focus on

expanding the model's application and evidence base, particularly through engagements with government organizations. We expect these experiences to provide critical insights into AI adoption under demanding operational, security, and governance conditions, further shaping both the model and the broader practice of sustainable AI transformation.

We extend our deepest gratitude to everyone who contributed to enable this work to happen and hope it serves as an exemplar to other similar collaborative initiatives.

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# Foreword by Anita Carleton

As organizations across every sector race to harness the transformative potential of artificial intelligence (AI), one truth becomes increasingly clear: the future of AI depends on its software engineering foundation. At the Carnegie Mellon® University (CMU) Software Engineering Institute (SEI), we have spent decades helping government and industry build that foundation—advancing the practices, architectures, and assurance methods that make complex software systems dependable, scalable, and secure. That same legacy now anchors our approach to AI adoption.

The SEI's history in developing maturity models, beginning with the Capability Maturity Model (CMM) and evolving into the CMM Integration (CMMI) framework, fundamentally changed how organizations assess and improve their engineering capabilities. These models gave industries a structured way to measure readiness, reduce risk, and continuously improve. Today, as AI technology moves from experimentation into mission-critical environments, organizations need similar clarity to understand where they are, where they need to go, and how to get there responsibly.

This AI Adoption Model builds on that heritage. It reflects the SEI's deep experience helping organizations adopt emerging technologies safely and effectively, and it incorporates the latest insights from our ongoing research in building trustworthy, secure, and engineering-grade AI. Across the SEI, our researchers are advancing methods for robust and resilient machine learning (ML), designing architectures for human-AI teaming, exploring scalable evaluation and assurance techniques, and addressing the unique risks posed by generative AI. These efforts aim to ensure that AI systems behave predictably, can be rigorously tested, and can be integrated into complex systems with confidence.

But technology alone is not enough. Successful AI adoption requires organizational readiness, responsible governance, and a commitment to continuous learning. It demands that leaders create environments where innovation is balanced with safety, where experimentation is guided by engineering discipline, and where AI's power is matched by our capacity to steward it wisely. The AI Adoption Model provides a practical pathway to achieve that balance—helping teams assess their current state, identify capability gaps, and mature their practices in a deliberate and sustainable way.

As we stand at a moment of unprecedented possibility, the organizations that thrive will be those that pair the creativity of AI with the rigor of software engineering. The SEI has long been a partner in enabling that rigor, and we remain committed to guiding the nation—and the world—through the next chapter of technological transformation.

I am pleased to introduce this AI Adoption Maturity Model and the thoughtful work behind it. I hope it serves as a valuable tool for leaders, practitioners, and policymakers working to realize AI's full potential while upholding the highest standards of quality, security, and trustworthiness.

**Anita D. Carleton**  
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# Foreword by Rajendra Prasad

At Accenture, we continuously strive to be the most comprehensively AI-enabled organization in the world. That ambition reflects not only the scale of our commitment, but also our conviction in AI's transformative potential.

Artificial intelligence is rapidly redefining what organizations can achieve. It is reshaping how industries operate, compete, and create value. It is a force as consequential as the industrial and digital revolutions combined, one capable of reinventing every part of the enterprise, from products and operations to talent and decision-making.

Organizations that successfully harness emerging technologies like AI will not merely improve how they work. They will fundamentally reinvent what they can become.

Over the past decade, Accenture has invested heavily to expand our AI capabilities, deepen our research, and embed AI across our own enterprise. Through this journey, we have learned that successful AI adoption requires far more than deploying advanced models or experimenting with new tools. It demands reinvention across strategy, data, operating models, governance, talent, and culture. Technology alone does not create value. Lasting impact comes from aligning people, processes, and platforms around a shared vision for transformation.

That is why we are honored to share the AI Adoption Maturity Model developed in collaboration with the Carnegie Mellon University Software Engineering Institute. At a moment when many organizations struggle to move from experimentation to enterprise-wide impact, this model provides leaders with something urgently needed: a rigorous yet practical assessment framework for understanding where they stand today, what capabilities they must strengthen, and how they can scale AI with confidence.

Our collaboration with CMU SEI combines deep academic and engineering expertise with the practical lessons we have learned through thousands of AI transformation engagements across industries. One insight stands out above all others: organizations that succeed with AI approach it systematically. They build maturity not only in technology, but in trust, governance, operational readiness, and human capability.

The AI Adoption Maturity Model provides a structured pathway for doing exactly that. It helps organizations move beyond isolated pilots toward scalable, trusted, enterprise-wide adoption. At a time when the pace of technological change continues to accelerate, the ability to scale responsibly and decisively will increasingly separate leaders from laggards.

This is a pivotal moment for every organization. The companies that define the future will be those that combine bold ambition with disciplined execution and responsible leadership. My hope is that this model serves as both a guide and a catalyst, helping leaders navigate complexity, accelerate reinvention, and create enduring value in the age of AI.

**Rajendra Prasad**  
Group Chief Executive  
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## Foreword by Kim Krogh Andersen

At Telstra, we sit at a unique intersection of the AI story. As one of the world's leading telecommunications companies, we are a provider of the digital infrastructure that AI-enabled services depend on. We are also a partner in the AI ecosystem, and a player who is navigating the complexity of embedding AI into our own operations, products, and customer experiences. That unique experience has shaped everything I have come to believe about what it really takes to adopt AI well.

In my role leading Network, Product, and Technology at Telstra, I am accountable for the network infrastructure, platforms, engineering systems, and products that serve millions of customers every day. Over recent years, we have matured from encouraging AI experimentation and asking “What’s possible?” to asking “What’s responsible, repeatable, and genuinely valuable?” It has been one of the most challenging but also energizing strategic transitions of my career.

What has made the transition hard is not the technology itself. The models are powerful and the tooling has matured rapidly. What trips organizations up—time and again—is the gap between technical ambition and operational reality. Successful AI adoption requires organizational discipline, clear accountability, and a relentless focus on value creation. It also requires strong investment and focus on workforce readiness and culture. The challenge to overcome is not a technology one; it’s a shortfall in organizational capability.

That is why the AI Adoption Maturity Model, developed by Carnegie Mellon University’s Software Engineering Institute in partnership with Accenture, resonates so strongly with me. It is not a framework that celebrates AI novelty—it is one that demands AI discipline. The five levels: Exploratory, Implemented, Aligned, Scaled, and Future Ready map directly onto the conversations I have every week with my teams: “How does this fit into our whole-of-business strategy? Are we building on our AI foundations to support reuse and scaling? How do we support the culture and adoption rates we need to truly see value?”

What distinguishes this model is its insistence on breadth. You cannot achieve AI maturity through engineering excellence alone. You need data readiness, workforce capability, responsible AI governance, and leadership alignment—all advancing together. The model forces

that holistic view, which in my experience separates organizations making enduring progress from those endlessly cycling through proofs of concept.

For technology and product leaders, I want to underscore one message in particular: to be successful, AI can’t be a technology strategy, it needs to be a whole-of-business strategy. Decisions about where AI should operate, what trade-offs are acceptable, and what success looks like are whole-of-business decisions. Use this framework as a bridge to bring your counterparts from across the business into the AI maturity conversation.

The telecommunications sector has long faced a trade-off between meeting rising customer expectations for coverage and reliability, and operating efficiently. AI presents a new opportunity to manage that trifecta of needs—if used well. And there will be similar dynamics playing out across other industries, requiring urgent action. But moving fast without getting the technical, governance, workforce and data foundations right first will slow organizations down in the long run.

If you are a technology executive, a product leader, or a board member trying to understand where your organization truly stands—lean into this framework. Use it to diagnose honestly, plan deliberately, and progress steadily. The goal is not to reach a particular level on a scale; it is to build an organization where AI delivers durable, trustworthy value. This model will help you get there.

**Kim Krogh Andersen**

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## Executive Summary

Fast-evolving advances in artificial intelligence (AI), particularly in generative AI technologies and agentic AI implementations, have challenged organizations to rethink how they conduct business. The drive to capitalize on AI for transformational productivity, cost improvements, and accelerated innovation continues to intensify. Yet many organizations are still struggling to realize the promised benefits of their AI initiatives, especially as they incorporate generative AI into products and workflows. Demonstrating value is particularly challenging as the scope of AI adoption increases, inundating all aspects of engineering, product development, and organization management. Lessons learned point to common challenges: mismatched expectations, misaligned applications, and poorly executed or untested implementation practices. These challenges highlight a persistent gap: both industry and the Department of War (DoW) lack an adaptable, measurement-based tool for consistently assessing AI adoption maturity, enabling predictable outcomes, and building a roadmap to strengthen the practices required to meet AI goals.

To address this need, the Carnegie Mellon® University (CMU) Software Engineering Institute (SEI), in partnership with Accenture, developed the AI Adoption Maturity Model v1.0. With the full model now available, organizations finally have a comprehensive, research-backed, and tested framework to guide their AI adoption journeys with confidence. This document introduces the model's key concepts and explains how organizational leaders can use it to overcome the challenges that arise as they work to realize the promise of AI. The model enables organizations to measure the degree to which essential practices are implemented and governed, ensuring they can adapt and deliver AI solutions that meet business needs with reliable and trustworthy outcomes. Its purpose is to drive the creation of a clear, actionable roadmap based on an assessment of organizational maturity.

The development of the AI Adoption Maturity Model was grounded in a disciplined, evidence-driven process based on extensive research, including executive interviews, a systematic review of more than a hundred existing AI maturity efforts worldwide, pilots of AI projects, an extensive industry survey, the CMU SEI's deep expertise in maturity modeling, and Accenture's global experience with AI implementation. The comprehensive review of AI maturity frameworks included sources



from academic literature, government reports, and industry efforts and excluded marketing products and incomplete models through a structured filtering process. The reviewed sources included maturity models and instruments which had the elements of maturity models. In total, the development team examined 56 sources in depth and identified key gaps across the landscape—most notably the lack of measurable criteria, limited adaptability to rapid AI advances, and inconsistent definitions of required practices. These findings directly shaped the AI Adoption Maturity Model's design, ensuring that it is actionable and can be aligned with AI capabilities as they evolve beyond generative and agentic AI.

This AI Adoption Maturity Model also builds on the foundation of the SEI's decades of experience developing some of the world's most influential maturity frameworks, including the Capability Maturity Model (CMM), Capability Maturity Model Integration (CMMI), CERT® Resilience Management Model (CERT-RMM), and Cybersecurity Maturity Model Certification (CMMC). These models have helped organizations in software, systems, cybersecurity, and service delivery achieve predictable performance and institutionalized process discipline. This model extends this proven track record into an environment shaped by rapidly evolving AI technologies, increasing security and resilience pressures, and rising organizational risk.

In addition, the AI Adoption Maturity Model has been deployed with several Fortune 500 companies as part of an early-adopter program, where it delivered demonstrably strong results and proved its effectiveness in accelerating disciplined, enterprise-scale AI adoption. Outcomes show that results are being cited verbatim in boardrooms, directly shaping next-step priority decisions, and producing an immediate shift from organizational fragmented AI efforts and churn to a controlled, disciplined, and sharply targeted AI adoption strategy that fills the gaps identified by the model.

The model defines five levels of AI adoption maturity—Exploratory AI, Implemented AI, Aligned AI, Scaled AI, and Future Ready AI—and details eight core organizational change and system lifecycle engineering dimensions that underpin each. Each level describes the capabilities organizations must master to advance. Although AI amplifies many of the security and risk management challenges that are present in software development, the completed AI Adoption Maturity Model maintains a sharp focus on AI-relevant practices, reducing duplication with other efforts in software development, organizational development, cybersecurity, risk management, and digital transformation.

The AI Adoption Maturity Model is based on the fundamental idea that organizations must be able to do the following to achieve winning AI implementations:

- align their desired objectives for AI with their organizational goals and vision
- understand the current state of their capabilities as relevant to AI
- establish a list of required improvement and implementation actions, in order of priority, against their desired state
- identify dependencies across actions and their priorities
- commit resources and time to enable the successful execution of the roadmap

To provide value while taking into consideration these challenges, the AI Adoption Maturity Model does the following:

- focuses on reliable, trustworthy, and repeatable measurement and analysis to drive maturity in AI practice
- provides flexibility to create customized views tailored to different domains, organization types, sizes, and contexts
- targets development of a roadmap as a key outcome in addition to an assessment of the maturity of capabilities
- includes AI-relevant capability areas designed to help an organization adapt to changes in the technology landscape
- provides logical groupings of these key AI-relevant capability areas through dimensions

As the gap widens between AI's potential and realized value, the AI Adoption Maturity Model offers a decisive way forward. It enables organizations to align AI efforts with business priorities and achieve reliable, scalable success across the enterprise. By emphasizing reliable, evidence-based measurement, the model empowers organizations to realize sustained value from their AI investments and target the capabilities that will deliver the greatest impact.

With this first road-tested version of the full model now available to the public, organizations across sectors can define their roadmap for more consistent, responsible, and effective AI implementation. Widespread use of this model has the potential to elevate industry-wide capability and accelerate meaningful, measurable progress in AI adoption by guiding organizations toward common processes and practices relevant to AI implementations.

# Introduction

Artificial intelligence (AI), from machine learning (ML) to generative AI and fast evolving agentic systems, is transforming how organizations operate. As interest in trustworthy human–AI collaboration accelerates, many organizations face rising complexity. Organizations have come to realize that achieving value from AI requires more than developing or onboarding AI models and investing in compute. It requires a number of key elements to align data readiness, validation rigor, workflow re-engineering, human-in-the-lead design, and infrastructure with business goals. These demands increase cost and uncertainty, leaving many organizations struggling to meet their objectives and searching for clarity in AI adoption.

Many organizations, and even entire disciplines, find themselves at a turning point where AI, generative AI, and agentic AI applications are redefining workflows and challenging existing assumptions of what it means to execute tasks. The discipline of software engineering serves as a clear example: generative AI-powered tools are accelerating code generation, summarization, and even implementation (albeit with caveats) [\[Ozkaya 2023\]](#). As these technologies evolve, software-driven organizations are exploring how AI can boost efficiency, productivity, creativity, and value while reducing costs. At the same time, organizations such as OpenAI, Google, Microsoft, and Anthropic are developing frontier models and intensifying competition by rapidly releasing new versions with increasingly advanced capabilities. This brings both benefits and drawbacks to organizations looking to adopt AI. Anthropic's Claude Mythos released in April 2026 is a good example. Early versions of generative AI models released in 2023–2024 could barely complete simple cyber tasks, while this release of Claude Mythos, when given network access, could execute complex, multi-stage attacks on vulnerable networks and exploit those vulnerabilities autonomously.

By the time this text is published, yet more new capabilities of these generative AI models will have emerged. This pace of change demands more than traditional return-on-investment (ROI) analysis and value focus in AI adoption. It calls for a clear focus on business goals, disciplined execution, consistent implementation of core engineering practices, rigorous measurement, and continual improvement. Only by preparing and maintaining all of these aspects at a predictable level can an organization stay ahead of the fast-evolving AI innovations and their risks—meeting the accelerating expectations for innovation



and reinvention through AI while keeping risks firmly under control. This is where a maturity model brings the necessary clarity, structure, and consistency.

## 1.1 Common Barriers to Effective AI Adoption

The gap between the expected and realized value from AI remains significant. Failures rarely stem from the technology alone. More often, they arise from mismatched expectations, misaligned use cases, insufficient data capabilities, lack of user understanding, talent shortages, and underestimated required time commitments especially for establishing and following repeatable processes [Lewis 2021]; [Ryseff 2024].

Many organizations adopt AI reactively, attempting to keep pace with peers, without evaluating their processes or defining how AI supports measurable outcomes. They fail to ask what AI *should do* for the business, getting stuck in what AI can do as they race to bring vendor solutions into their ecosystem. The SEI and Accenture research shows that organizations frequently lack the ability to scope or baseline their AI efforts, making it difficult to assess progress, prioritize initiatives, and mature key practices. Technical debt [Kruchten 2019] in AI infrastructure compounds these challenges, and rushing deployments only increases costs and risks [Cisco 2025]. This challenge is borne out empirically: in January 2026, the SEI and Accenture Research conducted a survey of 600 professionals and found that a lack of appropriate data (35%), insufficient knowledgeable staff (34%), and integration with legacy systems (32%) were the top factors negatively impacting AI adoption, underscoring that the barriers organizations face are not unique but reflect a systemic readiness deficit across the industry.

The SEI also interviewed more than two dozen executives responsible for leading AI initiatives. Across these discussions, three challenges consistently emerged:

1. uncertainty about what to baseline and measure to track improvement during AI adoption
2. difficulty scaling AI solutions effectively across the enterprise
3. difficulty keeping up with evolving technology and aligning it with software engineering practices and cybersecurity risk

To deliver value and ROI amid these challenges, organizations must identify the right workflows to envision with AI, identify key indicators to measure and assess progress, establish the capabilities

needed to measure success, and strengthen the skills required to scale solutions enterprise-wide or through their desired organizational scope. Establishing these goals will clarify engineering priorities, identify where AI-related technology investments will have the greatest impact, and ensure that all AI initiatives stay aligned with business objectives, sound software engineering practices, and strong cybersecurity risk management.

A structured, reliable guide is essential.

## 1.2 Building a Controlled, Measurable Pathway for AI Adoption

An important first step in resolving AI adoption challenges is to treat adoption as a rigorously managed process toward institutionalization—one that can be systematically controlled, measured, and optimized through clearly defined intermediary objectives. As with any mature process discipline, whether in manufacturing or software engineering, successful AI adoption requires executing a structured sequence of activities that reliably delivers the intended outcomes. When these activities are carried out with consistency and evaluated against well-specified outcome measures, organizations can establish repeatable, scalable practices essential for sustained, enterprise-level reliable and trustworthy AI adoption.

Effective AI adoption needs to start with understanding existing processes, identifying obstacles, and making decisions about where AI can accelerate business outcomes. Whether through onboarding commercial off-the-shelf software or building custom AI solutions, organizations need processes to guide AI implementations.

Measurement is a foundational pillar of this process and is precisely what distinguishes a rigorous maturity model from simple checklists, high-level practice inventories, or subjective self-assessments. An effective maturity model embeds objective, repeatable metrics that allow organizations to quantify capability, track progress, and validate outcomes—ensuring that AI adoption advances in a disciplined, evidence-driven, and repeatable manner. Yet current practice falls well short of this standard: the January 2026 SEI Accenture survey found that only 31% of organizations rely on business value and ROI prioritization when assessing and developing their AI adoption roadmap, suggesting that most organizations still lack the structured, measurement-driven approach this model is designed to provide.

To deliver value and ROI amid these challenges, organizations must identify the right workflows to envision with AI, identify key indicators to measure and assess progress, establish the capabilities needed to measure success, and strengthen the skills required to scale solutions enterprise-wide or through their desired organizational scope.

Effective AI adoption needs to start with understanding existing processes, identifying obstacles, and making decisions about where AI can accelerate business outcomes.

Measuring maturity through the AI Adoption Maturity Model establishes where an organizational unit stands and how well key practices are being executed.

There are two key aspects of measurement in the AI Adoption Maturity Model:

1. **Maturity measurement** is achieved by using the AI Adoption Maturity Model and assessing the degree to which an organizational unit meets goals, follows key practices, and accomplishes maturity indicators.
2. **Measurement and analysis practices** are embedded in the maturity model as key goals and supporting practices to provide guidance for collecting evidence. These measurement and analysis practices provide direction and increase the likelihood of success of activities, such as the implementation of AI use cases, re-engineering of workflows, transparency, explainability and responsible AI guidelines, or workforce development initiatives to name a few AI adoption related critical capabilities.

Together, these two forms of measurement ensure that AI adoption progresses in a disciplined and verifiable manner. Measuring maturity through the AI Adoption Maturity Model establishes where an organizational unit stands and how well key practices are being executed. The embedded measurement and analysis practices then provide the evidence needed to guide implementation activities towards ROI and value.

### 1.3 AI as a Software Engineering Capability

AI-enabled systems—and the AI capabilities now embedded in most commercial software products and services—still share many of the core requirements of traditional software systems. However, AI also introduces additional technical complexities—such as data-driven behavior, model lifecycle management, nondeterministic and inconsistent outputs, and increased dependency on continuous monitoring—that organizations must be prepared to address [Ozkaya 2021].

Predictive and generative AI models carry inherent uncertainty, increasing the importance of data management, validation, responsible AI practices, and security. Agentic AI adds further complexity, especially in metadata management and multi-system coordination, which necessitate managing orchestrated interactions among autonomous components, humans, and tools continuously. Robust system-level engineering is necessary to ensure safety, observability, and resilience and identifying the new failure modes that agentic systems exhibit.

Activities for verifying, validating, and securing AI systems need to account for ambiguity as well as an increased attack surface due to frequently changing data and the underlying nature of AI models.

Amid these complexities, foundational software engineering and cybersecurity practices remain essential. Organizations must design, test, integrate, and maintain AI components with the same discipline they apply to traditional systems, and also account for the new complexities introduced by these components. Key considerations include

- designing system architectures that account for uncertainty in data, AI models, and outputs
- ensuring user experiences make AI behavior transparent and more predictable
- managing differing rhythms of change across data, AI models, and business needs
- verifying and securing AI systems in the face of increased ambiguity and attack surface
- establishing robust monitoring capabilities to support measurement- and evidence-based decision making and timely failure management
- establishing processes to manage the creation, usage, monitoring, and evolution of new development artifacts indicated by generative and agentic AI applications, such as prompts, guardrails, and dynamic metadata

AI-enabled and agentic systems seem to behave like conventional systems—until they do not. These inflection points require clarity and structured guidance. The AI Adoption Maturity Model identifies the practices organizations must strengthen to bridge traditional engineering methods with the unique demands of AI.

Organizations must design, test, integrate, and maintain AI components with the same discipline they apply to traditional systems, and also account for the new complexities introduced by these components.



# 2

## Rapidly Shifting Focus of AI Adoption

## Rapidly Shifting Focus of AI Adoption

A major barrier to effective AI adoption today is the challenge of making sense out of the pervasive AI-washing that has saturated nearly every sector—software, telecommunications, transportation, healthcare, automotive, avionics, finance, marketing, and even small local enterprises. Today, many organizations pursue AI adoption with a vague, “AI everywhere” mindset rather than a clearly defined strategy. When multiple AI initiatives with different objectives overlap without a clear business direction, it becomes difficult to prioritize value assessment, the supporting practices, and resources needed. A critical first step is to define what AI adoption means for both the organization and the specific organizational unit. This may involve multiple goals—and potentially multiple maturity assessments—but intentionally separating them enables clearer focus and greater success. The AI Adoption Maturity Model reinforces disciplined scope management to support this outcome.

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Today, many organizations pursue AI adoption with a vague, “AI everywhere” mindset rather than a clearly defined strategy.

### 2.1 What Does AI Adoption Mean for You

AI adoption can take many forms, each representing a meaningful step in bringing AI capabilities into an organizational unit or the broader enterprise. These activities may include

- implementing vendor solutions that are fundamentally AI-driven (e.g., AI-native software-development integrated development environments [IDEs] or AI-enabled dictation tools)
- upgrading traditional vendor tools to new AI-enhanced versions (e.g., automated meeting summaries or action-item extraction features)
- re-implementing high-value business specific use cases with AI-augmentation
- redesigning end-to-end workflows to embed AI capabilities as integral components of the process
- deploying AI platforms or tools that impact multiple workflows within a major business function (e.g., marketing, software engineering, or contact-center operations)
- making an enterprise-level decision to adopt AI broadly, ensuring the workforce is equipped, enabled, and upskilled to use AI wherever it adds value (e.g., making off-the-shelf frontier models available for general use)

For most organizations today, the lines between these choices have effectively disappeared. This shift has been driven by the rapid expansion of generative AI services, tools, and agentic capabilities that can be integrated into software ecosystems and organizational enterprise functions with unprecedented ease.

This ease of bringing tools and services onboard should not be confused with successful AI adoption. AI adoption is a business choice driven by a business need, with a set of goals that can be effectively addressed by AI solutions. Successful use of a tool by everyone in the organization (e.g., any generative-AI-supported chat) is simply a tool roll out.

As AI capabilities proliferate and frontier models continue to advance, organizations are beginning to encounter two major emerging challenges:

1. rigorously evaluating the *cost of platform lock-in*, the ongoing operational cost of maintaining and integrating AI capabilities, and the true total cost of ownership associated with enterprise-scale AI adoption
2. *managing a rapidly evolving risk and security landscape* in which threat surfaces, assurance requirements, and governance expectations are continually shifting

Furthermore, organizations must confront the reality that AI systems are increasingly used to develop, optimize, or govern other AI systems. In this layered and highly dynamic landscape, it is critical for organizations to define scope with precision and map their capabilities directly to their AI adoption objectives. The AI Adoption Maturity Model is designed with explicit awareness of these evolving tiers of AI use. Organizations that determine their target maturity levels and institutionalize the associated practices will be able to apply those practices consistently across diverse AI initiative scopes and efficiently reuse them as new AI capabilities and use cases emerge.

Organizations need to start thinking about AI initiatives across two critical aspects:

1. the degree of use and autonomy of AI during creation and development
2. the level of AI embodied in the resulting product or workflow

The degree of AI use in creation captures how independently AI operate in generating solutions—ranging from traditional to AI-assisted to augmented, semi-autonomous, or potentially fully autonomous

creation. In the resulting workflow or product, AI technology progresses similarly, from traditional human-driven products to AI-enabled, AI-orchestrated, and autonomous systems (with a human in the lead in all cases). Together, these two aspects reveal a rapidly emerging shift in the technological landscape: organizations are entering a space where AI agents are increasingly included in designing, deploying, and managing AI-enabled workflows and products. As a result, the distinction between the tools organizations use (AI as creator) and the systems they build (AI within the product) is dissolving, underscoring the essential need for new governance, architectural rigor, and measurement discipline that maturity approaches must be cognizant of. [Figure 1](#) summarizes these forces to help organizations think through their AI adoption priorities.

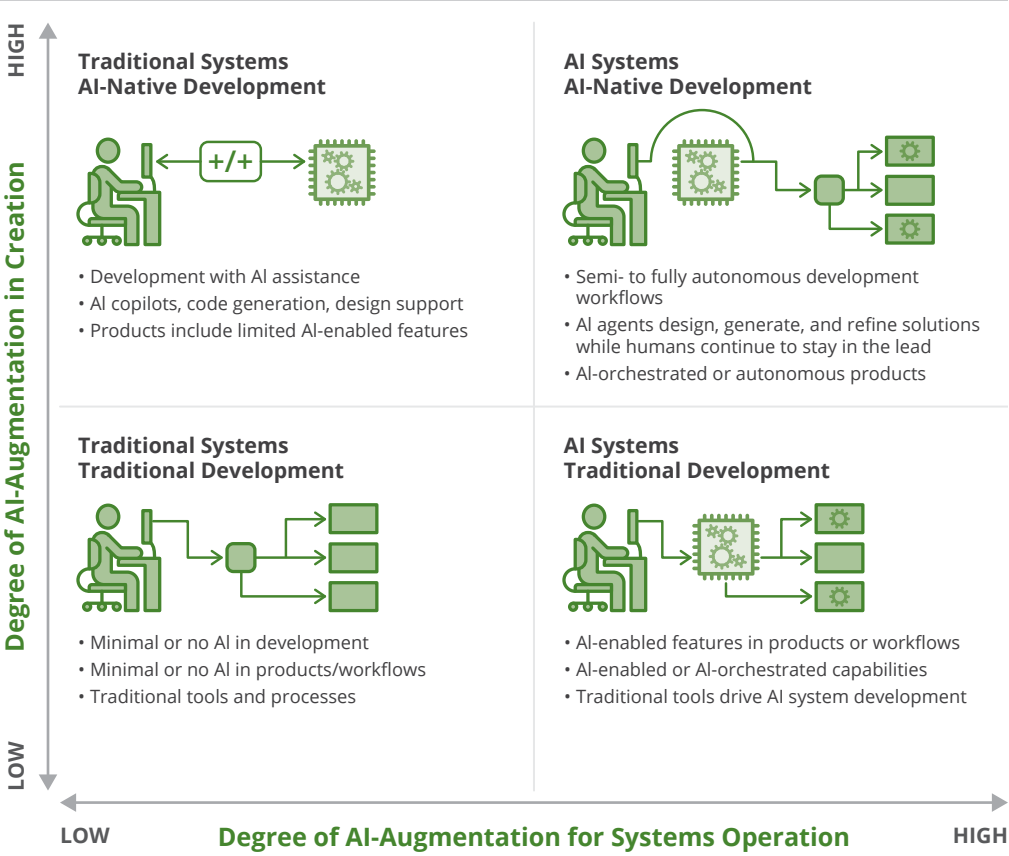


Figure 1: AI Adoption Complexities



The growing pervasiveness of AI makes it essential for organizations to carefully consider their *organizational scope* before beginning any assessment or process-improvement initiative aimed at enhancing AI adoption outcomes.

## 2.2 Assessment Scope

When planning an AI adoption assessment, deciding on the scope is a critical first step: successful AI implementations focus on data and workflows that pertain to a particular business unit or service. Judicious scoping of AI Adoption Maturity Model assessments will also help stakeholders interpret assessment results since they will be relevant to a specific domain and include actionable roadmaps to drive improvements.

The assessment sponsor determines the scope by selecting the specific group or component of the organization to assess (the “organizational unit”[OU]) and the specific domains and capability areas of the AI Adoption Maturity Model to examine. These are *organizational scoping* and *model scoping*, respectively. Together they guide the assessment boundaries and next step actions.

The growing pervasiveness of AI makes it essential for organizations to carefully consider their *organizational scope* before beginning any assessment or process-improvement initiative aimed at enhancing AI adoption outcomes. Organizational leaders frequently desire assessments that cover the entire enterprise, and, in that situation, the organizational unit is simply “the enterprise.” For an assessment focused on one component of an enterprise, for example a subsidiary or a business unit, the organizational unit is that subsidiary or business unit.

Within the organizational unit there may be a single implementation of AI focusing on a critical workflow that spans multiple business units in the organization (e.g., patient intake), several implementations, or a function that applies to the entire organizational unit (e.g., human resources or software engineering).

The AI Adoption Maturity Model includes areas considered essential to safely adopting AI and achieving predictable ROI. However, organizations may choose to use the AI Adoption Maturity Model based on their own criteria; the model contains *context attributes* that allow users to make intentional decisions about the model scope. For example, organizations that are not subject to specific regulatory obligations may choose to de-scope or defer assessment against areas related to compliance. [Section 4.1](#) describes context attributes and model scoping in further detail.

While there are many benefits of enterprise-wide assessments, narrowing the scope to a specific organizational unit enables more targeted insights and actionable outcomes. This is particularly true because successful AI use is very focused on data and workflows that pertain to a particular business unit or service. The benefits of a more focused assessment scope include results that center on areas deemed to be more “AI ready” at the start of the organization’s journey, and which may be of particular relevance to staff in those areas. Judicious scoping also helps organizations to avoid falling into the trap of treating single use cases or vendor roll outs as AI adoption.

The first element of this model asks whether an AI adoption scope is clearly defined and consistent with the organization’s mission, vision, and business objectives to help organizations focus on the value AI adoption will bring and to ensure the assessment can provide an actionable roadmap.

While there are many benefits of enterprise-wide assessments, narrowing the scope to a specific organizational unit enables more targeted insights and actionable outcomes.

# 3

## Levels of AI Adoption Maturity

## Levels of AI Adoption Maturity

The AI Adoption Maturity Model helps an organization determine the level of AI maturity to pursue and guide the management of its implementation roadmap. Levels are an abstraction useful for executives and represent maturity with specific dimensions and capability areas; they align with an overall AI strategy and drive actionable prioritizations for a roadmap.

The AI Adoption Maturity Model characterizes the AI adoption journey into one of five maturity steps to support an organization's goals. [Figure 2](#) shows the five levels of AI adoption maturity.

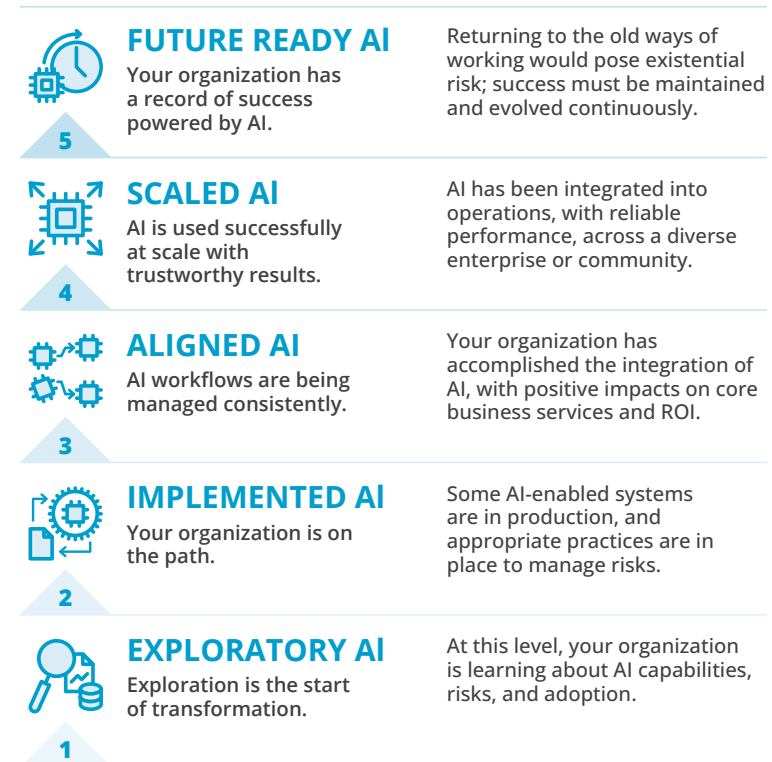


Figure 2: The Five Levels of AI Adoption Maturity

As discussed in [Section 2.2](#), the “organizational unit” (the part of the organization being assessed for maturity) can be scoped to include an entire organization or a single component. The term “organization” refers to the larger enterprise context.



Developing a roadmap for organization-wide AI adoption requires a wider assessment scope—often involving multiple assessments across the enterprise—and is driven by different factors than a more limited, initial approach to AI adoption.

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The five levels of maturity are as follows:

1. **Exploratory AI:** At this level, the organizational unit is learning about AI capabilities with regard to the identified scope, risks, and adoption. The goal at this level is to ground AI adoption in the organizational unit's context, culture, and objectives. Exploration is the start of transformation, with a limited set of use cases to establish minimal standard practices. AI-enabled systems and workflows may exist in both sandbox and non-reported production environments.
2. **Implemented AI:** The organizational unit is implementing AI-enabled systems and workflows, resulting in some AI-enabled systems and workflows in production with potential ROI and value. At this level, priorities include making sure baseline delivery, data, and security-related practices are in place to manage risks. The organizational unit is on the path for successful AI adoption, demonstrating the realization of some use cases.
3. **Aligned AI:** At this level the organizational unit has accomplished the process of AI integration with potential positive impacts on core business services and ROI. Appropriate workflows are identified and managed consistently with AI applications. It is clear how the organizational unit can utilize AI and how AI maps to business goals.
4. **Scaled AI:** At this level, AI has been integrated into operations, with reliable performance. The organizational unit and the organization can implement AI initiatives at scale, apply AI to new use cases and scopes with ease, and ensure reliable implementations and trustworthy outcomes.
5. **Future Ready AI:** The organizational unit has a track record of success powered by AI. The organizational unit can consistently deploy and scale AI initiatives and seeks to maintain and evolve its success with AI continuously. Team members empowered by AI successfully drive innovation for the organizational unit.

These levels of maturity were selected after reviewing the challenges and successes that organizations are realizing; studying the existing software, AI, and cybersecurity maturity landscape; and empirically collecting

experiences from the field. One key insight these efforts revealed is that AI adoption maturity, even at low levels of maturity, needs to have an intentional focus. Organizations should explicitly define their purpose for adopting AI and the corresponding maturity level that aligns with that purpose. An intentional decision to achieve a lower maturity level, such as Exploratory or Implemented AI, will inherently require fewer capability areas. As a result, gaps in the adoption of AI will be observable and frequent communication of the purpose for adopting AI across the organization will help to avoid the misidentification of deficiencies.

A common misconception about maturity models is that every organization should strive to reach the highest maturity level. In practice, the AI Adoption Maturity Model is not intended as a compliance ladder where all organizational units must progress sequentially to Level 5. Instead, the maturity levels represent different operating models and capability profiles that align with varying business objectives, risk tolerances, investment strategies, and scopes of AI adoption.

An organizational unit should deliberately determine its target maturity level based on the value it seeks to achieve from AI. For example, a business unit that uses AI selectively to improve internal productivity may find that the Aligned AI maturity level provides the capabilities necessary to achieve its goals and manage associated risks. In contrast, an organization whose products, services, or competitive advantage depend heavily on AI may require the capabilities associated with the Scaled AI or Future Ready AI maturity level.

The maturity levels therefore describe an increasing ability to adopt, operationalize, govern, and evolve AI, but they should not be interpreted as mandatory stages of advancement. Some organizational units may intentionally remain at a lower maturity level because additional investments would provide limited business value. Others may choose to accelerate capabilities in specific areas while maintaining lower maturity in others, depending on their strategic priorities.

Viewed this way, maturity becomes a strategic decision rather than a compliance exercise. The purpose of the model is to help organizations understand their current capabilities, identify gaps relative to their desired outcomes, and make informed investment decisions about where greater maturity will create meaningful value. The most effective maturity target is not necessarily the highest level—it is the level that best aligns AI capabilities, governance, engineering practices, and organizational objectives.

An organization may experiment with AI in fragmented ways, but true maturity emerges when those efforts are aligned, managed, and sustained at scale.

Ad hoc, inconsistent AI capabilities—or gaps in practices and measurement—are common in early adoption, but they do not define maturity. Maturity is determined by the presence of intentional, repeatable, and governed approaches: clear objectives, standardized practices, measurable outcomes, and continuous improvement. An organization may experiment with AI in fragmented ways, but true maturity emerges when those efforts are aligned, managed, and sustained at scale. As such, the AI Adoption Maturity Model is developed to drive success at any level by staging and recommending key capability areas to invest in. Advancing in maturity, therefore, is achieved through realizing the key capabilities, across the eight dimensions, essential for each respective level, as defined in [Section 4](#). In addition, advancing maturity for each dimension implies realizing the maturity indicators ([Section 4.13](#)), which include key attributes of established capability area maturity, such as resource allocation, relevant governance, and evidence of defining and managing key activities.

A key benefit of the AI Adoption Maturity Model is that it helps organizations institutionalize effective practices by translating broad AI ambitions into a structured set of capabilities that can be systematically implemented, measured, and improved over time. The model establishes a common language for discussing expectations, priorities, risks, and outcomes across business, technology, governance, and operational stakeholders.

By defining a practical progression path, the model helps organizations determine which capabilities should be established first and which can be developed as adoption matures. It transforms implicit, inconsistently applied activities into explicit organizational practices, reducing dependence on individual expertise and increasing repeatability across teams and initiatives. Most importantly, the AI Adoption Maturity Model shifts the conversation from “What AI technology should we adopt?” to “What organizational capabilities must we develop to consistently realize value from AI?” In doing so, it serves not merely as a measurement framework, but as a mechanism for building organizational discipline, accelerating learning, and creating the conditions for sustainable AI adoption at scale. While many factors influence maturity transitions, the core objective is to build reliable and trustworthy capabilities that enable organizations to achieve their desired business outcomes by using measurable practices and adapting to rapidly evolving AI technologies.

### 3.1 Maturity Level 1: Exploratory AI



Exploratory AI maturity, the first level of AI adoption maturity, is defined by experimentation and is the first step to discovering where AI should be applied. Exploration is how many organizations intuitively reacted when the first versions of generative AI models were introduced. Several organizations responded, and continue to respond, by defining use cases where generative AI could be applied, starting sandbox initiatives to see if they could implement simple use cases, or holding hackathons to expand the aperture of use cases to select from. The same process unfolded when agentic AI implementations accelerated.

At the Exploratory AI level, the organizational unit is assessing use cases and workflows for AI applicability and resources for feasibility. The organizational unit may not have an AI strategy beyond a goal to understand where AI could be effective in the organization. A key goal is learning from experimentation. The unit can achieve the Exploratory AI level by focusing on two key capability areas: [Experimentation](#) and [AI-Literate Workforce Development](#).

Key actions, therefore, to accomplish the Exploratory AI level of maturity include the following:

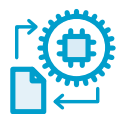
- Establish an ability to develop, select from, and assess use cases and workflows. The [Experimentation](#) capability area emphasizes an organizational unit's ability to design, conduct, evaluate, and learn from controlled AI technology pilot studies. An essential component of achieving the Exploratory AI level is learning how to scope and resource an AI project in a reliable way. While many organizations know to start with experimentation, they often do not include an explicit validation step for comparative and predictable outcomes. The scale of this challenge is striking: 61% of organizations in the SEI and Accenture Research January 2026 industry survey report having a formal AI strategy that has not been completely implemented, a finding that suggests the majority of enterprises are effectively stranded between the Exploratory and Implemented AI levels. They possess strategic intent but lack the disciplined execution practices this model is designed to build. Effective exploration sets the foundation for effective maturity improvement. In addition, given the pace of change in the AI technology landscape, experimentation is a capability that the organizations need on an ongoing basis.

Beginning workforce development at the initial level of AI adoption maturity is an intentional choice given today's reality of AI evolution.

- Develop skills and training capability through an AI-literate workforce. [AI-Literate Workforce Development](#) is an essential component for AI adoption and a reflection of the reality of AI technology.

Beginning workforce development at the initial level of AI adoption maturity is an intentional choice given today's reality of AI evolution. Generative AI and agentic AI implementations, more so than other forms of AI, come with their own opportunities and challenges. The inherent uncertainty in the structure of the AI model, the variation of correct outcomes, the potential for a larger security attack surface, the use of unstructured data, and the organization's data being potentially compromised means basic skills and training must be available from the start. Today many routine tools considered necessary for business are being revised to include generative AI-based features in addition to other forms of AI components. Across the board, organizations that report AI successes have access to talent. On the flip side, access to skillsets is among the top challenges for organizations in their AI adoption experience. Building skills and training out of the gate are an essential component of Exploratory AI and also lay the foundation for increasing maturity.

### 3.2 Maturity Level 2: Implemented AI



An organizational unit at the Implemented AI maturity level deploys core AI-enabled features, internally for organizational use or externally to its customers, even at a limited scope. Organizational units targeting Implemented

AI deploy AI-enabled systems, workflows, and products in production, involving production data or solutions that the organizational unit is using to support business processes. Examples of drivers that enable maturity at the level of Implemented AI are characterized by the following:

- [AI Strategy Development](#) is a priority.
- Select workflows are deployed and have achieved operational relevance. This will require key capability areas related to data to be developed: [Data Lifecycle Management](#) and [Data Quality Assurance](#).
- The ability to deliver AI-enabled features, even at a limited scope, requires engineering competencies, bringing the [AI Architecting](#) and [Test and Evaluation](#) capabilities to the forefront.

- AI implementations are known and understood more broadly and are governed. Risk is managed appropriately and relevant [Responsible AI](#), [Policy and Compliance](#), and [Risk Management](#) practices are in place.

The following are key actions to implement to ensure successful Implemented AI maturity and establish a path forward to effectively focus on the innovation and workflow re-engineering required to achieve the third level of maturity, Aligned AI.

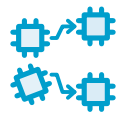
- The organizational unit will need to establish an AI strategy. AI adoption should align with organizational values and principles, ensuring that goals are communicated clearly, stakeholders are aware of roles and responsibilities, and the AI strategy effort is driven by business priorities, not the fear of “missing out” on AI. Given the proliferation of AI to all aspects of the business and software ecosystem, an AI strategy to manage the risks and business expectations is not a choice but a necessity. The AI strategy can be part of other business strategy initiatives or can be stand alone. This is an organizational choice.
- The organizational unit will need to master all aspects of data management and governance. Established organizations will likely have existing practices around data management that can be extended for relevance to AI. However, AI adoption, especially in generative AI and agentic AI implementations, creates new requirements for managing data, establishing pipelines for preparing data, establishing governance around data, and managing security aspects. For example, key practices for unstructured data need to be in place. The organizational unit needs to have mastered the management of metadata and dynamic data dependencies as agentic AI implementations take priority.
- Similarly, capabilities around [AI Model Management](#) need to be established. A core challenge with AI adoption is keeping up with the pace of evolution. Organizations can avoid wasting resources unnecessarily when new versions of AI models are released by defining process and practices to drive model evolution. This will reduce degraded outcomes due to declining model performance. It is common for system outcomes to diminish when the performance of newer model versions degrades. Organizational units will need to establish controlled assessment practices to make decisions with an empirical basis. As frontier model capabilities continue to evolve, many organizations are reaching decision points where they are reliant on the support of multiple frontier models.



This shift creates a need for robust AI model management practices early in the maturity journey to address complexities such as vendor lock-in and dependencies between models.

An organizational unit which has achieved the Implemented AI level across relevant dimensions demonstrates successful AI-enabled system and workflow deployment and is ready to build on this foundation to achieve organizational transformation through AI.

### 3.3 Maturity Level 3: Aligned AI



The Aligned AI level is characterized by workflow re-engineering and a strategic focus on innovation.

At this level, the organizational unit has achieved the key practices to develop, monitor, and resource AI initiatives across business units and other functions. The organizational unit has fundamental business practices in place (e.g., workforce, data, governance, risk management, and engineering capabilities) to reliably challenge existing approaches, either within its internal processes or with designed and delivered products, to take advantage of AI by aligning AI initiatives along with other tools. The [Measurement and Analysis](#) capability and supporting practices are established for assessing the success of use cases as well as the success of AI adoption, as guided by the AI strategy.

While at the Aligned AI level of maturity, organizational units are realizing changed workflows and establishing key capabilities to achieve repeatable success, they have not yet demonstrated the reliability and trustworthiness of all capabilities at scale. The Aligned AI level of maturity is where organizations make their primary infrastructure and business investments, building on established measurement, analysis, and monitoring practices to support informed decision-making.

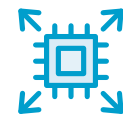
Key actions to take to achieve the Aligned AI level of maturity include the following:

- Demonstrate competency in the [Workflow Re-Engineering](#) dimension.
- Establish goals and practices for the [Organizational Culture Evolution](#) capability area, addressing all the capabilities needed to support [Workforce and Culture](#).
- Ensure core processes and practices for managing AI risk are supported by defined processes, including [Human-in-the-Lead Automation](#) and [Transparency and Explainability](#).

- Building on the established AI strategy, drive [AI Partnership Development](#) to support the vision of AI initiatives and AI adoption goals which will be the foundation for success at scale.

Maturity indicators for each dimension will need to demonstrate an objective evaluation of policy adherence, descriptions of how exceptions are addressed, high-level management involvement and ownership of risks, resource allocations, and a commitment to removing barriers to adoption. The organizational unit needs to demonstrate these key strategic capability areas and maturity indicators to position itself for the next level, Scaled AI maturity.

### 3.4 Maturity Level 4: Scaled AI



As an organizational unit moves through the Exploratory AI, Implemented AI, and Aligned AI maturity levels, implementing key capability areas and pursuing maturity indicators of governed, measured, and predictable practices for each applicable dimension, it must set itself on the right path to achieve reliable, trustworthy, and scalable AI initiatives. A complexity that AI adoption brings is the increasing need to manage the supply chain and AI ecosystems through partnerships.

While an organization at the Aligned AI level will have already demonstrated how to select partners and assess their AI capabilities against the organization's expectations, a key indicator of success at the Scaled AI level is the ability to objectively define what AI capabilities are needed to scale AI across the organizational unit. As the organizational unit will have already established mechanisms to identify and allocate resources and responsibilities, the focus of Scaled AI will be replicating these initiatives by employing the knowledge gained and focusing on shared resources to optimize costs.

Key maturity signals to advance to the next level, Future Ready AI, include the following:

- Establish the [Deployment at Scale](#) capability area where the business, technical, and infrastructure requirements are aligned to support AI-initiatives across the organization.
- Strategic decisions include focused discussions on how AI can support increased value to both the organization and to its workforce.

- Shared artifacts define the success of AI initiatives; this goes beyond those limited to select AI functions, which may have been the case when the organization was at the Implemented AI or Aligned AI stages.
- At the Scaled AI level, organizations have also established optimal [AI-Literate Workforce Development](#) initiatives and are gaining benefits from these initiatives, including sharing talent, tools, procedures, and practices across initiatives.

### 3.5 Maturity Level 5: Future Ready AI



Achieving the Future Ready AI level means that the organization, or organizational unit, has made the intentional decision to be AI-native where AI is embedded in the organization's culture, the organization knows how to take advantage of old and new advances rapidly, and the organization continues to innovate with AI while augmenting innovations with other relevant tools and approaches. For example, Accenture is an organization that has made the intentional and clear decision to be a Future Ready AI company and has clearly communicated this with its workforce, investors, customers, and partners. While many organizations will adopt AI, not all will make the decision to be Future Ready. This is a strategic business decision that requires aligning AI capabilities with organizational objectives, as well as assessing the value and resources necessary to support those objectives.

A Future Ready AI organization can assess new AI innovations for their relevance, usefulness, and timeliness and compare them to the existing AI initiatives with concrete criteria. These organizations employ a predictive capability to more accurately forecast the implications of AI innovations for enhanced organizational performance and business objectives. The organization has foundational capabilities and processes and practices in place to resource new initiatives and evolve them as needed.

Organizations or organizational units at the Future Ready AI maturity level have achieved the following:

- Improvements are identified proactively using established analytic techniques (e.g., causal analysis, statistical methods, and simulations). The organizational unit will need to master predictive measurement and modeling, demonstrate the use of analytic models, and continue data collection and analysis to drive decision making and predictable planning.

- Existing processes and practices and analytic techniques are utilized to assess new technical advancements in the AI landscape and move initiatives quickly through the Exploratory AI, Implemented AI, Aligned AI, and Scaled AI levels, effectively and with planned and monitored outcomes.
- Community outreach initiatives are established that support and influence the AI practitioner, tooling, and research community in innovation.
- The workforce is clear about its responsibilities, knows where the resources are, and, when changes occur, is not perplexed and can rely on the established practices and analytic techniques to make informed assessments about the relevance of AI technology and its expected outcomes.

As an example, organizations today struggle to assess the risks and benefits of agentic AI and where to employ it. Agentic AI is not a new form of AI—though it is often mentioned in conjunction with the advancements of generative AI and frontier models. Agentic AI brings together elements in the AI landscape to develop goal-oriented, adaptive, and self-corrective capabilities to support the development of AI-enabled systems by delegating decision making to autonomous AI systems.

A Future Ready AI organization would be able to rely on its established practices in [Experimentation](#) to determine use cases relevant to applying agentic AI, rely on existing [Responsible AI](#) and [Human-in-the-Lead Automation](#) practices to assess governance risks, and borrow skills and practices from [AI Architecting](#) to incorporate agentic AI into its system design decision tradeoffs. At the Future Ready AI level, the organization has the required capabilities and maturity indicators to avoid surprises with data and anticipate future changes.

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# 4

## The AI Adoption Maturity Model

### The AI Adoption Maturity Model

This section describes the AI Adoption Maturity Model in detail, beginning with the basic structural elements of the model and their relationships, as shown in [Figure 3](#).

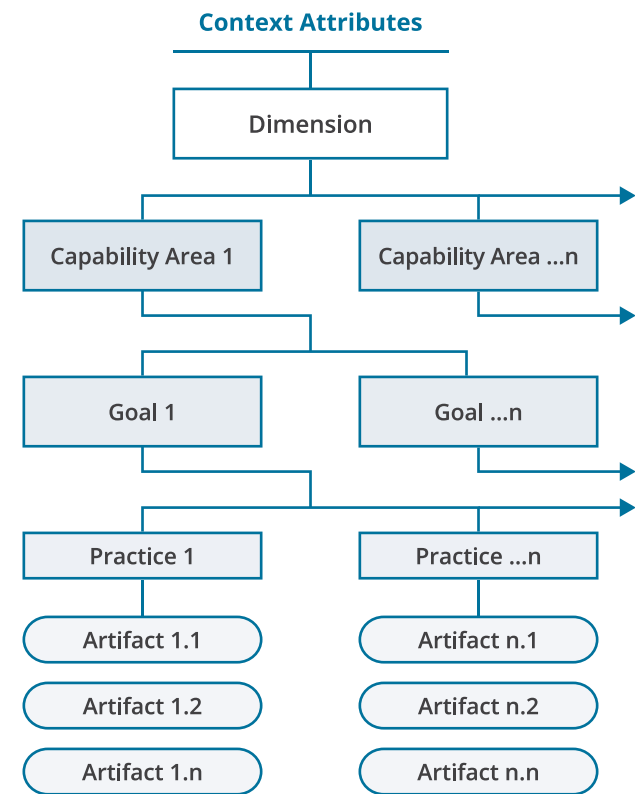


Figure 3: Building Blocks of the AI Adoption Maturity Model

These building blocks are defined as follows:

- **Context Attributes**—elements that characterize the organizational unit being assessed based on its organizational, technology, and AI functions to most appropriately scope the assessment against its current and target capability maturity
- **Dimensions**—the core foundational pillars grouping capabilities for AI adoption and maturity



- **Capability Areas**—logical groupings of related goals that together improve an organization’s performance in specific skills and activities
- **Goals**—high-level statements of the outcome to be achieved by the effective implementation of a group of practices within a capability area
- **Practices**—activities that are considered important to achieve a goal
- **Maturity Indicators**—verifiable attributes, metrics, or observable characteristics to assess the extent to which a capability area satisfies the criteria associated with a defined maturity level. Maturity indicators provide the objective evidence of institutionalization, performance, and consistency, enhancing the reliable evaluation of progression across maturity levels.
- **Artifacts**—tangible evidence demonstrating that specific capabilities, goals, practices, or processes are established and consistently performed. Example artifacts may include policies, metrics, reports, tool outputs, architecture documentation, training materials, or operational data.

#### 4.1 Context Attributes

The realities of AI adoption demand a maturity approach that is both rigorous and adaptive—one that supports a rapid pace, structural flexibility, and operational efficiency in both assessment and implementation. Organizations must be able to evaluate capability maturity continuously, respond to evolving technologies and risks, and translate insights into incremental, value-driven progress without sacrificing governance or engineering discipline. Accordingly, an AI adoption maturity model should focus on a concise set of core capability areas while remaining flexible enough to evolve with changing technologies, emerging best practices, and varying organizational contexts.

The AI Adoption Maturity Model accomplishes these goals through a set of context attributes to perform model scoping as introduced in [Section 2.2](#). A number of context attributes focused on organizational functions, technology functions, and AI functions help scope the assessment and subsequent roadmap activity. [Table 1](#) summarizes some of these attributes.

Context attributes provide flexibility in the use of the model, guiding the scope of the capability areas, their relevant goals, and practices to assess against to steer the organization to those capabilities that matter most to

Table 1: Representative Context Attributes

|                          | ATTRIBUTES                      | DESCRIPTION                                                                                                                         | EXAMPLE VARIABLES                                                                             |
|--------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| ORGANIZATIONAL FUNCTIONS | Organization footprint          | The organizational unit’s operational footprint, which includes its physical locations, legal entities, and so on                   | National, Regional, Global                                                                    |
|                          | Domain                          | Area of operation, such as an organizational unit’s industry, products, or customer base                                            | Finance, health care, avionics, ...and so on                                                  |
|                          | Regulatory posture              | Extent and scope of rules, controls, and oversight that an organizational unit is subject to, usually from governmental regulations | Regulated, semi-regulated, unregulated                                                        |
|                          | Size                            | Number of staff relevant to the organizational unit                                                                                 | 100–1K, 1K–10K, 10K–100K, >100K                                                               |
|                          | Planning horizon                | Timeline for specific AI adoption goals for the organizational unit                                                                 | <6 months, 6 months – 1 year, 1–3 years, >3 years                                             |
|                          | Organizational scope            | Boundary of the AI adoption maturity discussion                                                                                     | Team, business unit, business sector, program, organization                                   |
| TECHNOLOGY FUNCTIONS     | Software engineering strategy   | The extent to which software development supports the organizational unit’s business goals                                          | Software organization, only IT functions, software outsourced                                 |
|                          | Core technology competency      | Structure of the scope and strategy of the organizational unit’s competencies                                                       | Product first, AI first, single core-competency, multi-industry                               |
|                          | Data readiness                  | Data preparation and sources for AI                                                                                                 | Datasets are AI-ready, datasets are partially AI-ready, datasets are purchased for AI, hybrid |
| AI FUNCTIONS             | Primary AI development approach | The extent to which the organizational unit will develop AI initiatives                                                             | Build, buy, customize, hybrid                                                                 |
|                          | AI risk profile                 | The degree of risk the organizational unit is comfortable assuming                                                                  | High, medium, low                                                                             |
|                          | AI business goal                | The archetype that characterizes the organizational unit’s approach to AI adoption                                                  | Thought leader or AI innovator, early adopter, early majority, late majority, laggard         |
|                          | AI staffing                     | The organizational unit’s approach to staffing AI initiatives                                                                       | In-house, contracted, none                                                                    |

The AI Adoption Maturity Model is designed to evolve as the technology landscape and organizational scope change.

align AI initiatives with its goals. Moreover, these attributes introduce adaptability in applying the AI Adoption Maturity Model and support empirically grounded variations in the improvement strategies organizations may pursue.

Context attributes, for example, can help decouple several AI adoption scopes. Understanding an organization’s AI development approach is one way of scoping with context attributes. This knowledge forecasts how fully the organization will pursue its AI initiatives and thus helps in determining what capabilities are most essential to its context and which capabilities it might outsource.

An organization that aims to fine-tune and customize AI models will need different competencies than one that will use solutions out of the box. An organization that aims to build its agentic AI systems in-house will need different competencies than one that will have other organizations build these systems for them.

The AI Adoption Maturity Model is designed to evolve as the technology landscape and organizational scope change. The notion of context attributes is one mechanism used to accomplish this by focusing on the capability areas that are most relevant to the characteristics of the organization. In addition, the context attributes give the maturity assessors a mechanism to include other relevant practices based on the characteristics of the organizational and technology landscape as the use of the AI Adoption Maturity Model expands.

One example of a scenario where context attributes can help is when two organizational units are collaborating on AI implementations and they need to be aware of their maturity in relevant dimensions and prepare for gaps. These context attributes can provide the organizational units with an initial look at similarities and differences and guide the prioritization of dimensions and capability areas.

### 4.2 Eight Core Dimensions of Capability

Dimensions are the core foundational aspect of AI adoption and maturity and group the capability areas of the AI Adoption Maturity Model. Each dimension, through its capability areas, also includes maturity indicators of characteristics that demonstrate maturity in relation to implementing the goals through the example practices. The AI Adoption Maturity Model includes eight dimensions around two key high-level goals: (1) organizational change and (2) AI lifecycle engineering, as shown in [Figure 4](#).

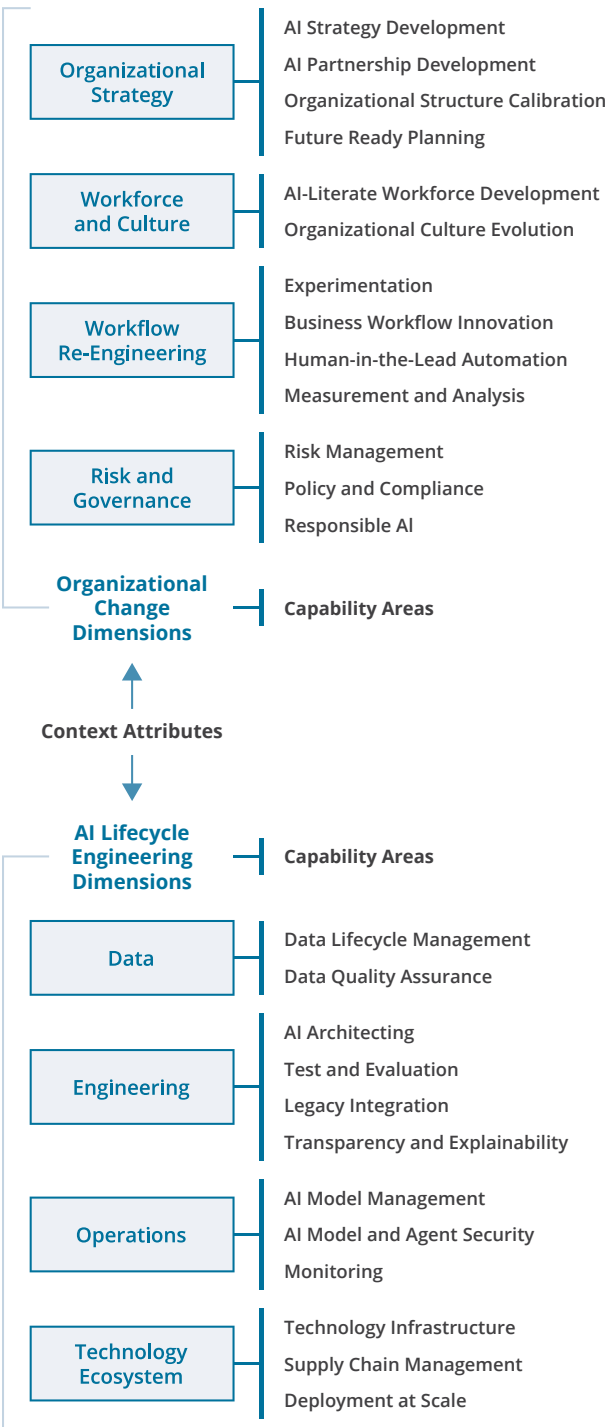


Figure 4: Eight Core Dimensions and Corresponding Capability Areas

To sustain an organization's focus on AI adoption, the content of each of these capability areas is focused on identifying AI-relevant practices only. Although there may be merit and efficiencies to be gained from a broader focus on more general aspects of process maturity across software and cybersecurity, those should be appraised through other successful models like CMMI or CMMC. Similarly, when the goal is to assess AI risk exposure of a specific system or service, existing approaches like NIST AI Risk Management Framework [NIST 2023] can be used. It is expected that an organization which already has mastered aspects of maturity in software and cybersecurity will approach AI adoption with the same level of rigor and maturity mindset.

#### 4.2.1 Organizational Change Dimensions

AI adoption is an organizational change initiative. Successful AI adoption necessitates the structured transformation of an organization's strategies, operating models, processes, technologies, workforce capabilities, and cultural norms to achieve improved performance or adapt to evolving internal or external conditions. The four Organizational Change dimensions, described in detail below, focus on business, workforce, workflow, and risk-related capability areas.

**Organizational Strategy** focuses on how effectively an organization positions itself to deliver business value through AI, whether its adoption is limited in scope or integrated broadly across the enterprise. It includes the capability areas which focus on [AI Strategy Development](#), [AI Partnership Development](#), [Organizational Structure Calibration](#), and [Future Ready Planning](#).

**Workforce and Culture** focuses on capability areas to grow an AI-literate organization through its people, including the training and development that they will need to succeed in integrating, developing, and/or using AI solutions appropriately. It includes the capability areas which focus on [AI-Literate Workforce Development](#) and [Organizational Culture Evolution](#).

**Workflow Re-Engineering** emphasizes the changes in workflows and the experimentation that will be needed to use AI to transform key business practices; assist, augment, and autonomously support people's work as needed; and take proper advantage of AI in the business, including following sound measurement and analysis practices. This dimension is core to innovation with AI and includes the capability areas which focus on [Experimentation](#), [Business Workflow Innovation](#), [Human-in-the-Lead Automation](#), and [Measurement and Analysis](#).

**Risk and Governance** focuses on capability areas that cover key standards, policies, and practices to establish and follow to manage AI and related risks from the perspective of the organization, technology, and end users. It includes the capability areas which focus on [Risk Management](#), [Policy and Compliance](#), and [Responsible AI](#).

#### 4.2.2 AI Lifecycle Engineering Dimensions

AI Lifecycle Engineering dimensions focus on the technical and engineering aspects of AI adoption related to AI-enabled systems and workflows. As is the case with the Organizational Change dimensions, system AI Lifecycle Engineering dimensions build on existing software engineering and cybersecurity practices.

**Data** focuses on maintaining relevant, high-quality, and secure data practices that directly strengthen the value and alignment of AI initiatives. This dimension includes the capability areas which focus on [Data Lifecycle Management](#) and [Data Quality Assurance](#).

**Engineering** focuses on the key capabilities required to transform AI use cases into production-ready solutions, including integrating, developing, and testing AI components and solutions. This dimension includes the capability areas which focus on [AI Architecting](#), [Test and Evaluation](#), [Legacy Integration](#), and [Transparency and Explainability](#).

**Operations** brings together capability areas essential for collecting production data, monitoring systems for success, and establishing feedback loops for effective long-term sustainment and evolution. This dimension includes the capability areas which focus on [AI Model Management](#), underscoring secure operations through [AI Model and Agent Security](#), and [Monitoring](#) to enable reliable, value-added AI adoption.

**Technology Ecosystem** focuses on the capabilities needed to properly use and keep up to date on AI technologies and required infrastructure to ensure the efficiency and scalability of AI solutions. This dimension includes the capability areas which focus on [Technology Infrastructure](#), [Supply Chain Management](#), and [Deployment at Scale](#).



### 4.3 Navigating the AI Adoption Maturity Model

Sections 4.4–4.11 present each dimension, its capability areas, example goals, practices, and artifacts. Each capability area includes

- that capability area's relevant goals and practices
- the maturity level required to achieve that capability area (Exploratory AI, Implemented AI, Aligned AI, Scaled AI, or Future Ready AI)
- example artifacts to help guide the documentation, consistent execution, and process maturity for that capability area

Note that example artifacts are intended only as examples and their actual implementation is likely to follow other organization-specific processes and guidelines. It is also important to recognize that there is no one-to-one mapping of artifacts and other evidence of maturity; this is typically a many-to-many matrix where one artifact cuts across several capability areas, and each capability area is served by several artifacts with one or two being core artifacts.

Each dimension discussion concludes by highlighting related capability areas from other dimensions, illustrating the interconnections between capability areas and dimensions. The model can be navigated by maturity level, dimension, or capability area. Organizations may use it to identify a target maturity level, assess current capabilities, explore related practices and artifacts, and/or understand dependencies across dimensions. This flexible structure supports both focused improvement efforts and enterprise-wide AI adoption planning.

### 4.4 Organizational Strategy Dimension

The Organizational Strategy dimension examines how effectively an organization positions itself to deliver business value through AI for the selected organizational scope.

#### 4.4.1 AI Strategy Development Capability Area

*The ability of the organizational unit to define how AI will support desired business outcomes through a realistic, evolvable, and executable AI strategy*

##### Goals and Practices

|                   |                                                                                                                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 1</b>     | An AI adoption scope which is consistent with the organization's mission, vision, and business objectives is clearly defined for the organizational unit. Processes and practices are in place to |
|                   | <b>Practice 1</b> Identify and involve stakeholders relevant to the possible scope for AI adoption.                                                                                               |
|                   | <b>Practice 2</b> Identify the business goals and requirements that are relevant to AI adoption.                                                                                                  |
|                   | <b>Practice 3</b> Articulate AI adoption goals clearly with objective and quantifiable outcomes.                                                                                                  |
| <b>Goal 2</b>     | <b>Practice 4</b> Align adoption with overall business objectives, values, and principles, including safety, human-centered implementation, and responsible use.                                  |
|                   | A strategy for adopting AI within the relevant scope is established. Processes and practices are in place to                                                                                      |
|                   | <b>Practice 1</b> Identify key goals, evaluation criteria, and milestones for the AI strategy.                                                                                                    |
|                   | <b>Practice 2</b> Document the AI strategy.                                                                                                                                                       |
| <b>Goal 3</b>     | <b>Practice 3</b> Communicate the strategy for AI adoption, key milestones, and success criteria to impacted stakeholders.                                                                        |
|                   | <b>Practice 4</b> Monitor performance and progress against the strategy (e.g., as part of existing governance activities).                                                                        |
|                   | AI adoption progress is regularly assessed against business goals, values, and principles in line with the AI strategy. Processes and practices are in place to                                   |
| <b>Practice 1</b> | Establish a process to assess and share AI adoption progress against values, goals, and principles.                                                                                               |

|                   |                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Practice 2</b> | Periodically review the AI strategy to ensure that it is current, relevant, and adjusted based on positive or negative outcomes.                           |
| <b>Practice 3</b> | Use the outcomes of the analysis to correct alignment of AI initiative and strategy with organizational business goals, values, and principles, as needed. |
| <b>Practice 4</b> | Evolve AI strategy by establishing follow-on actions as AI initiatives demonstrate positive or negative outcomes.                                          |



All the goals and applicable practices of the AI Strategy Development capability area need to be accomplished for this dimension to be at **Implemented AI Level Maturity (Level 2)**.

Example Artifacts

- AI strategy documentation
- section of a larger organizational strategy which includes AI adoption strategy, mission and vision documents
- documentation of business goals that will be measurably improved with AI
- communication plan
- examples of internal newsletters communicating AI adoption related progress
- project plan of AI adoption with milestones
- value stream mapping

4.4.2 AI Partnership Development Capability Area

*The ability to select and continuously evaluate partners for alignment with the organizational and AI strategy*

Goals and Practices

|                   |                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 1</b>     | Awareness of the evolving AI industry and related ecosystems is maintained. Processes and practices are in place to       |
| <b>Practice 1</b> | Establish AI-specific requirements for partners and third parties that align to organizational culture and risk appetite. |
| <b>Practice 2</b> | Regularly assess AI market trends, new versions, and new technologies to inform decisions.                                |
| <b>Practice 3</b> | Regularly assess the evolution of other technologies that support AI in the technology stack for relevance.               |

|                   |                                                                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Practice 4</b> | Identify potential partners and their capabilities in the organization's industry or community of interest.                                                                                                       |
| <b>Practice 5</b> | Review and communicate the state of the industry and potential partners to internal stakeholders.                                                                                                                 |
| <b>Goal 2</b>     | Potential external partners, including vendors, whose AI capabilities are consistent with AI adoption goals, are engaged. Processes and practices are in place to                                                 |
| <b>Practice 1</b> | Identify the acquisition approach for AI data, models, and solutions (e.g., buy off-the-shelf, fine-tune existing solutions, or build in-house).                                                                  |
| <b>Practice 2</b> | Identify key external technology enablers whose capabilities complement rather than replicate existing AI capabilities.                                                                                           |
| <b>Practice 3</b> | Evaluate prospective AI vendor capabilities (e.g., regarding data sharing, security, responsible AI, human-centeredness, suitability, and evolution approaches against fitness to AI strategy and risk appetite). |
| <b>Practice 4</b> | Define licensing requirements and other costs as they relate to AI solutions.                                                                                                                                     |
| <b>Practice 5</b> | Select third-party vendors and services based on their ability to meet scaling objectives and other requirements.                                                                                                 |

|                   |                                                                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 3</b>     | Contracts and mechanisms to form relationships are reviewed to ensure consistency with AI adoption goals. Processes and practices are in place to      |
| <b>Practice 1</b> | Form contracts and partnerships that preserve flexibility and control (e.g., avoiding undesirable vendor lock-in).                                     |
| <b>Practice 2</b> | Define ownership of data, models, overall systems, system artifacts, and so on in contracts.                                                           |
| <b>Practice 3</b> | Define performance expectations, quality and acceptance criteria, and measurements in the contract for partners and externally developed AI solutions. |
| <b>Practice 4</b> | Include vendor non-performance or failure of performance expectations as applicable in contract language concerning breach and remediation.            |



All the goals and applicable practices of the AI Partnership Development capability area need to be accomplished for this dimension to be at **Aligned AI Level Maturity (Level 3)**.

Example Artifacts

- market research documents, such as competitive assessment
- market analysis
- trend analysis
- evaluation matrix
- gap analysis
- AI-relevant checklists for contracts

4.4.3 Organizational Structure Calibration Capability Area

*The ability for the organizational unit to examine their current organizational processes and structures and align them to support AI adoption*

Goals and Practices

|        |                                                                                                                                                                                                                                                |                                                                                                                                        |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | Organizational processes are assessed to identify necessary changes or to define new processes required to effectively to manage AI adoption and AI initiatives. Processes and practices are in place to                                       |                                                                                                                                        |
|        | Practice 1                                                                                                                                                                                                                                     | Identify the organizational and technology processes that are relevant to AI adoption.                                                 |
|        | Practice 2                                                                                                                                                                                                                                     | Review organizational processes and AI strategy alignment regularly to identify changing business needs.                               |
|        | Practice 3                                                                                                                                                                                                                                     | Add or change organizational processes based on the review.                                                                            |
| Goal 2 | Practice 4                                                                                                                                                                                                                                     | Evaluate and update as needed change management processes to manage the impacts of AI technology, solutions, and initiatives.          |
|        | The organizational structure is proactively assessed to ensure support for AI adoption, including—as relevant—design, development, implementation, monitoring, and operationalization of AI solutions. Processes and practices are in place to |                                                                                                                                        |
|        | Practice 1                                                                                                                                                                                                                                     | Determine personnel, partners, and workflows associated with the organization's AI initiatives.                                        |
|        | Practice 2                                                                                                                                                                                                                                     | Map ownership and responsibility for personnel, partners, and workflows to the organizational structure.                               |
|        | Practice 3                                                                                                                                                                                                                                     | Review the effectiveness of the organizational structure in executing AI initiatives and achieving AI-relevant business goals and ROI. |
|        | Practice 4                                                                                                                                                                                                                                     | Revise the organizational structure to improve progress on AI adoption.                                                                |

- All the goals and applicable practices of the Organizational Structure Calibration capability area need to be accomplished for this dimension to be at **Aligned AI Level Maturity** (Level 3).

Example Artifacts

- documents which share guidance that defines an organization's core purpose and long-term aspirations—what they expect to achieve and how progress will be measured
- organizational structure which calls out AI adoption roles
- AI strategy
- 360 Framework
- enumeration of core functions leading to AI adoption

4.4.4 Future Ready Planning Capability Area

*The ability to prepare for possible futures by using analytic models (e.g., causal, statistical, and simulation) and maintaining a program of outreach and collaboration to construct strategic roadmaps for AI-related capabilities (e.g., skills, processes, and technologies)*

Goals and Practices

|        |                                                                                                                                                                               |                                                                                                                                                                                                     |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | Analytic methods for planning, monitoring, developing, testing, and deploying AI-related capabilities are established and maintained. Processes and practices are in place to |                                                                                                                                                                                                     |
|        | Practice 1                                                                                                                                                                    | Establish and maintain analytic methods (e.g., causal, statistical, simulation) to measure the effectiveness of the organizational unit's AI-related capabilities in achieving business objectives. |
|        | Practice 2                                                                                                                                                                    | Use analytic methods and models to monitor the business-as-usual use of the organizational unit's AI-related technology and predict (estimate) outcomes and failure rates.                          |
|        | Practice 3                                                                                                                                                                    | Use analytic methods to determine where the organizational unit's AI-related capabilities should be improved and evaluate proposed changes for impact on achieving business objectives.             |
|        | Practice 4                                                                                                                                                                    | Use analytic methods in root cause analysis, to identify how to prevent failures.                                                                                                                   |
|        | Practice 5                                                                                                                                                                    | Periodically validate the accuracy and performance of analytic models (e.g., by comparing predicted results to actual results over time).                                                           |



|        |                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                            |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 2 | The AI strategy is periodically updated to position the organization across multiple futures and time horizons. Processes and practices are in place to                                     |                                                                                                                                                                                                                                                                                                                            |
|        | Practice 1                                                                                                                                                                                  | Establish and maintain a set of possible futures (or scenarios) for the business environment, operational and organizational risks, and AI-related capabilities.                                                                                                                                                           |
|        | Practice 2                                                                                                                                                                                  | Elicit ideas for improving the organization's AI-related capabilities from internal sources (e.g., organizational brainstorming sessions, proposals volunteered by staff) and external sources (e.g., those engaged in technology scouting, outreach efforts, external collaborator discussions, and benchmarking visits). |
|        | Practice 3                                                                                                                                                                                  | Develop possible organizational roadmaps based on inputs.                                                                                                                                                                                                                                                                  |
|        | Practice 4                                                                                                                                                                                  | Use analytic models to evaluate alternative roadmaps, including an evaluation of estimated time and cost to implement.                                                                                                                                                                                                     |
| Goal 3 | Practice 5                                                                                                                                                                                  | Select roadmaps that provide resilience to unexpected events (changes to business environment, threat sources, and underlying technologies), and document as part of an updated AI Strategy.                                                                                                                               |
|        | AI related community outreach and collaborations are established to shape the relevant communities and support the organization's long-term vision. Processes and practices are in place to |                                                                                                                                                                                                                                                                                                                            |
|        | Practice 1                                                                                                                                                                                  | Develop collaborations with thought leader organizations (academia, industry, authorities).                                                                                                                                                                                                                                |
|        | Practice 2                                                                                                                                                                                  | Share best practices and experiences to improve and influence the community.                                                                                                                                                                                                                                               |
|        | Practice 3                                                                                                                                                                                  | Identify developments and future directions for possible selection and inclusion in the organization's AI strategy and capabilities.                                                                                                                                                                                       |



All the goals and applicable practices of the Future Ready Planning capability area need to be accomplished for this dimension to be at **Future Ready AI Level Maturity (Level 5)**.

Example Artifacts

- predictive analysis
- future scenarios
- future planning measurement framework using baseline or benchmark performance
- collaboration exemplars and case studies

4.4.5 Capability Areas Supporting the Organizational Strategy Dimension

Capability areas within the Organizational Strategy dimension span multiple maturity levels, reflecting the need for the organization's AI adoption strategy to evolve in sophistication, alignment, and scope as overall maturity increases. In addition, the following capability areas are closely related to Organizational Strategy:

- Refer to [Workforce and Culture: Organizational Culture Evolution](#) for goals and practices related to developing organizational capabilities.
- Refer to [Technology Ecosystem: Supply Chain Management](#) for goals and practices related to managing and maintaining existing third-party dependencies for vendor partners.
- Refer to [Risk and Governance: Risk Management](#) for goals and practices related to overall risk management around additional partnerships and procurement risks.

[Figure 5](#) presents an example assessment outcome for the Organizational Strategy dimension, showing the result of each capability area evaluated against maturity levels based on the extent to which the organizational unit meets the relevant goals, practices, and maturity indicators.

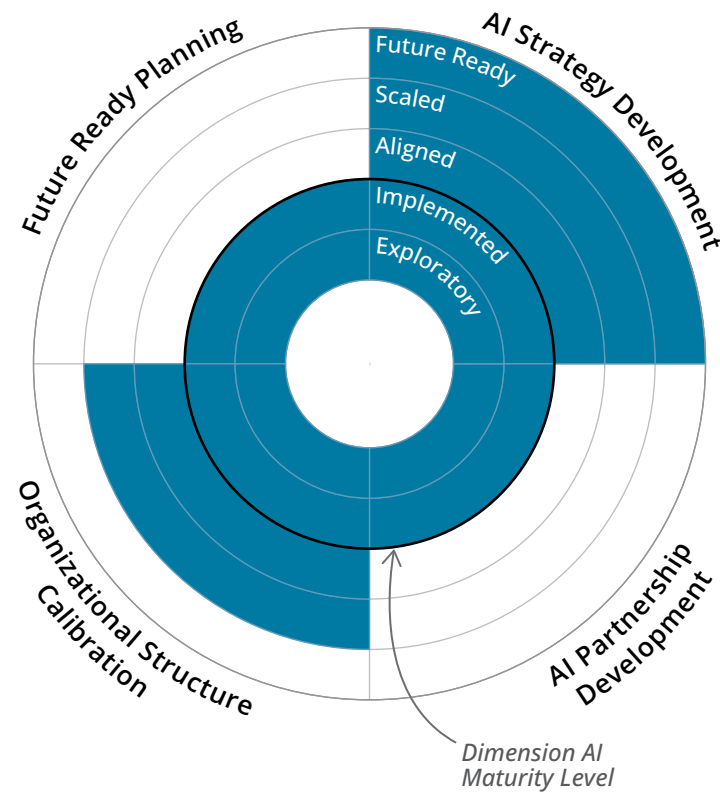


Figure 5: Example Assessment Outcome for the Organizational Strategy Dimension and Capability Areas

## 4.5 Workforce and Culture Dimension

The Workforce and Culture dimension focuses on capability areas required to build an AI-literate organizational by fostering an open-minded culture and providing the training and skills needed to effectively integrate, develop, and use AI solutions.

### 4.5.1 AI-Literate Workforce Development Capability Area

The ability to select and develop qualified staff with the skills and knowledge needed to effectively use AI within their work

#### Goals and Practices

|        |                                                                                                                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | Roles and competencies (knowledge, skills, and abilities) that support AI adoption are identified and documented. Processes and practices are in place to                             |
|        | <b>Practice 1</b> Identify the roles, competencies, and skills needed to support AI adoption within the organizational unit.                                                          |
|        | <b>Practice 2</b> Include completion of training and AI-related competencies in the organization's workforce management process and career progression ladder.                        |
| Goal 2 | <b>Practice 3</b> Benchmark staff skills to industry or community peers (informally or formally, such as through discussions with other entities or through a consulting engagement). |
|        | Training is delivered to develop and maintain AI-related competencies. Processes and practices are in place to                                                                        |
|        | <b>Practice 1</b> Establish training requirements and processes for AI-related roles.                                                                                                 |
|        | <b>Practice 2</b> Create or acquire AI-related training.                                                                                                                              |
|        | <b>Practice 3</b> Deliver training.                                                                                                                                                   |
|        | <b>Practice 4</b> Enforce compliance with training requirements.                                                                                                                      |

 All the goals and applicable practices of the AI-Literate Workforce Development capability area need to be accomplished for this dimension to be at **Exploratory AI Level Maturity (Level 1)**.

Example Artifacts

- knowledge, skills, and abilities assessments
- role and job family descriptions
- skills surveys analysis
- skills taxonomy or framework
- workforce assessment and gap analysis
- training documents and guidelines
- course outlines; training materials (training materials on the responsible use of AI)

4.5.2 Organizational Culture Evolution Capability Area

*The ability to establish a proactive, AI-native culture within the organizational unit through human resource processes and communication to change the mindset of how the organization operates*

Goals and Practices

|        |                                                                                                                                       |                                                                                                                                                                                                      |
|--------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | A culture that is open, participative, collaborative, and engaged in learning is established. Processes and practices are in place to |                                                                                                                                                                                                      |
|        | Practice 1                                                                                                                            | Establish and sustain an open, collaborative, and participatory work environment, where AI strategy is developed collaboratively and individuals are encouraged to experiment with AI.               |
|        | Practice 2                                                                                                                            | Communicate business values and principles to teams and individuals and reinforce AI adoption key messages regularly.                                                                                |
|        | Practice 3                                                                                                                            | Collect and analyze internal and external stakeholders' understanding and perceptions of AI initiatives, including those who will be affected by the outputs, actions, or decisions of AI solutions. |
| Goal 2 | Staff aligned with the organizational unit's AI culture are acquired and developed. Processes and practices are in place to           |                                                                                                                                                                                                      |
|        | Practice 1                                                                                                                            | Recruit, hire, and transition new staff (including from within the organization) capable of taking on key roles in current and near-future AI-related work.                                          |
|        | Practice 2                                                                                                                            | Recognize and reward individuals and teams that make contributions to the organizational unit's AI adoption goals and AI-native culture.                                                             |
|        | Practice 3                                                                                                                            | Enable staff to rapidly acquire the skills they need when they take on new AI-related roles.                                                                                                         |

|               |                                                                                                         |                                                                                                                                                                              |
|---------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <b>Practice 4</b>                                                                                       | Assess adoption strategy and performance to identify legacy or traditional jobs, competencies, or skill levels that may be deprecated or become less critical in the future. |
|               | <b>Practice 5</b>                                                                                       | Re-skill staff as necessary, to support AI adoption.                                                                                                                         |
| <b>Goal 3</b> | Workforce preparation for AI adoption is monitored and managed. Processes and practices are in place to |                                                                                                                                                                              |
|               | <b>Practice 1</b>                                                                                       | Track work performance on AI-related projects so that teams can identify relevant near-term skills development activities that should be pursued.                            |
|               | <b>Practice 2</b>                                                                                       | Include performance against and support for AI adoption in staff goals and objectives.                                                                                       |
|               | <b>Practice 3</b>                                                                                       | Revise or establish career tracks in particular areas of AI that are key to the organization's future work activities.                                                       |

 All the goals and applicable practices of the Organizational Culture Evolution capability area need to be accomplished for this dimension to be at **Aligned AI Level Maturity (Level 3)**.

Example Artifacts

- culture, skills, behaviors, and mindsets documentation
- cultural adoption metric
- AI adoption stakeholder communication artifacts (newsletters, town halls, external resources made available to the workforce)

4.5.3 Capability Areas Supporting the Workforce and Culture Dimension

Given the realities of AI infiltration to all aspects of system and product development as well as business operations, the [AI-Literate Workforce Development](#) capability area is expected to be accomplished at the **Exploratory AI** maturity level, while the [Organizational Culture Evolution](#) capability area will evolve as AI adoption maturity reaches **Aligned AI** level, closely related to [AI Strategy Development](#). In addition, the following capability areas are also closely related to Workforce and Culture.

- Refer to [Organizational Strategy: AI Strategy Development](#) and [Organizational Structure Calibration](#) for goals and practices related to the AI adoption direction which drive Workforce and Culture.



Figure 6 presents an example assessment outcome for the Workforce and Culture dimension, showing the result of each capability area evaluated against maturity levels based on the extent to which the organizational unit meets the relevant goals, practices, and maturity indicators.

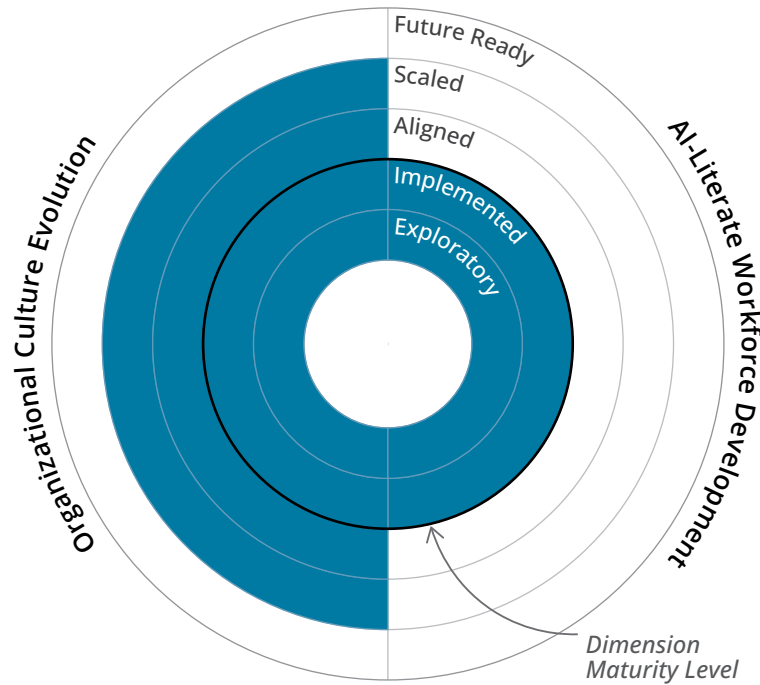


Figure 6: Example Assessment Outcome for the Workforce and Culture Dimension and Capability Areas

## 4.6 Workflow Re-Engineering Dimension

The Workflow Re-Engineering dimension focuses on analyzing existing business processes for potential improvements, innovations, and recommendations of new workflows. AI can be used to transform key business practices, to assist and augment people’s work, and to support business goals with sound measurement and analysis practices.

### 4.6.1 Experimentation Capability Area

The ability of an organizational unit to experiment with use cases and the application of AI to support the design, execution, and evaluation of AI-powered workflows

#### Goals and Practices

|        |                                                                                                                                                                                   |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | Proofs of concept are created to confirm expectations, engage stakeholders, and identify risks. Processes and practices are in place to                                           |
|        | <b>Practice 1</b> Identify business and workflow objectives and opportunities.                                                                                                    |
|        | <b>Practice 2</b> Identify use cases and proofs of concept.                                                                                                                       |
|        | <b>Practice 3</b> Define the minimal viable AI functionality in an iterative process.                                                                                             |
|        | <b>Practice 4</b> Compare the organization's compliance and other enterprise requirements against proofs of concept to ensure alignment.                                          |
| Goal 2 | <b>Practice 5</b> Define deployment readiness criteria, including performance, risk controls, and user readiness.                                                                 |
|        | Operational experimentation is conducted to validate proofs of concept, the underlying use cases, and their technical and business risks. Processes and practices are in place to |
|        | <b>Practice 1</b> Plan and communicate experiment design, including the people who will participate (end users), the tasks they will be asked to attempt, and success criteria.   |
|        | <b>Practice 2</b> Provide a representative sandbox area or environment to support the lifecycle of experiments, including testing and data collection.                            |

Practice 3

Collect qualitative and quantitative data about performance to refine and improve the proof of concept and use case.

Practice 4

Analyze experimentation findings to provide prioritized recommendations for operational deployment and scaling.



All the goals and applicable practices of the Experimentation capability area need to be accomplished for this dimension to be at **Exploratory AI Level Maturity (Level 1)**.

Example Artifacts

- experiment plans
- use case and workflow experimentation management dashboards
- resource plans for experimentation
- user experience (UX) and human-computer interaction (HCI) artifacts
- sandbox reference documents
- experiment results
- recommendations
- close out artifacts

4.6.2 Business Workflow Innovation Capability Area

*The ability of the organizational unit to rethink and redesign existing business processes and workflows to leverage AI with improved value-based outcomes and ROI*

Goals and Practices

Goal 1

Business processes and workflows that can be supported by AI to create efficiencies and improvements are identified and prioritized. Processes and practices are in place to

Practice 1

Identify and document workflows and business processes that align to the intended scope for AI adoption.

Practice 2

Analyze candidate business processes for efficiency (e.g., identify steps that do not add value, quantify drivers and inputs to cost, and review existing key performance indicators [KPIs] and historical performance).

Practice 3

Evaluate whether workflow performance objectives can be feasibly achieved through a non-AI process optimization before investing in AI-enabled solutions.

Practice 4

Prioritize candidate business processes and workflows for AI adoption according to estimated improvements and ROI against business goals.

Goal 2

Prioritized business workflows are assessed to identify risk and ensure the appropriateness of applying AI.

Practice 1

Identify the requirements and key performance indicators for prioritized business processes.

Practice 2

Identify the use cases and application instances of AI that support prioritized workflows.

Practice 3

Conduct technical reviews to evaluate the applicability and appropriateness of applying AI solutions to workflows, including assessing data volume, velocity, variety, veracity, value, and risks.

Goal 3

AI-enabled workflows are validated for operational scaling. Processes and practices are in place to

Practice 1

Define operational ownership and support responsibilities for AI-enabled workflows.

Practice 2

Validate integration with upstream and downstream systems with fallback, exception handling, and escalation mechanisms for AI failures.

Practice 3

For agentic AI workflows, clarify the handoffs and workflows involving agents and humans and the possible implications for organizational structure and staff responsibilities.

Practice 4

Ensure that periodically or based on changes, requirements and controls for AI solutions are updated based on the requirements and inherent risks of the workflows.



All the goals and applicable practices of the Business Workflow Innovation capability area need to be accomplished for this dimension to be at **Aligned AI Level Maturity (Level 3)**.

Example Artifacts

- business impact analysis (BIA)
- service-level agreement (SLA)
- recovery time objective (RTO)
- ROI analysis
- adoption rate
- business process documentation
- use case diagrams

### 4.6.3 Human-in-the-Lead Automation Capability Area

*The ability to design AI systems that ensure appropriate human accountability, oversight, and judgment in decisions made by or informed by the system's outputs—particularly when those decisions affect individual or community health, safety, quality of life, reputation, or autonomy*

#### Goals and Practices

|                   |                                                                                                                                                                                                                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 1</b>     | Systems and workflows are designed to empower humans by providing assistance, augmentation, or autonomy, as appropriate, through understandable interfaces and interactions. Processes and practices are in place to                                                                                                              |
| <b>Practice 1</b> | Design workflows and interfaces that are human-centered, empowering humans with appropriate decision oversight and accountability throughout the process.                                                                                                                                                                         |
| <b>Practice 2</b> | Clearly communicate the human and AI responsibility for outcomes and enforce ethical, equitable, and responsible development and use of AI.                                                                                                                                                                                       |
| <b>Practice 3</b> | Regularly collect and integrate into the system feedback from users regarding system usability, trustworthiness, correctness, and derived value regularly collected.                                                                                                                                                              |
| <b>Practice 4</b> | Conduct regular reviews of AI system performance and audit automated actions to identify issues that negatively affect human-in-the-lead automation.                                                                                                                                                                              |
| <b>Goal 2</b>     | Oversight for critical decisions is implemented. Processes and practices are in place to                                                                                                                                                                                                                                          |
| <b>Practice 1</b> | Define critical and irreversible decision points or processes that, based on risk, criticality, or anticipated AI performance, require enhanced, continuous, and/or specific human oversight. (Examples include outcomes affecting individual and community health, individual autonomy, safety, quality of life, or reputation.) |
| <b>Practice 2</b> | Define how an AI failure would impact the process or outcomes of critical and irreversible decisions.                                                                                                                                                                                                                             |
| <b>Practice 3</b> | Establish oversight and control mechanisms for workflows requiring human actions, at critical and irreversible decision points.                                                                                                                                                                                                   |

 All the goals and applicable practices of the Human-in-the-Lead Automation capability area need to be accomplished for this dimension to be at **Aligned AI Level Maturity** (Level 3).

#### Example Artifacts

- test plans
- subject matter expert (SME) assessments
- usability test findings and recommendations
- UX study plans and insights
- critical decision point analysis documentation with workflows
- tests of workflow execution based on AI risks

### 4.6.4 Measurement and Analysis Capability Area

*The ability to measure and analyze AI-enabled workflows and initiatives' effectiveness and performance*

#### Goals and Practices

|                   |                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 1</b>     | Measurement objectives, success criteria, and activities for AI-enabled workflows and business goals are identified. Processes and practices are in place to |
| <b>Practice 1</b> | Establish and maintain measurement objectives derived from the workflow goals and organizational unit objectives.                                            |
| <b>Practice 2</b> | Specify activities to measure the AI-native workflows against organizational unit objectives.                                                                |
| <b>Practice 3</b> | Obtain and store measurement data.                                                                                                                           |
| <b>Goal 2</b>     | Measurement results for AI-enabled workflows are provided. Processes and practices are in place to                                                           |
| <b>Practice 1</b> | Analyze and interpret collected measurement data.                                                                                                            |
| <b>Practice 2</b> | Determine the appropriate method of communicating results.                                                                                                   |
| <b>Practice 3</b> | Establish measurement dashboards to address project-specific and division needs and for the appropriate management levels.                                   |
| <b>Practice 4</b> | Define the frequency and methodology for future measurements.                                                                                                |

 All the goals and applicable practices of the Business Workflow Innovation capability area need to be accomplished for this dimension to be at **Aligned AI Level Maturity** (Level 3).



Example Artifacts

- benchmark and baseline performance
- dashboard with live tracking KPIs
- AI model performance (e.g., accuracy, F1 scores, speed/latency/throughput, response and recovery time)
- objectives and key results (OKRs)
- predictive analysis—success and failures
- rate of adoption
- reports to board members and shareholders
- stories about success and change
- template for test design and reporting
- test plan and expectations
- communication strategy and planning
- ROI analysis
- service continuity process
- service-level agreements

4.6.5 Capability Areas Supporting the Workflow Re-Engineering Dimension

The Workflow Re-Engineering capability areas need to be established for the **Aligned** AI level of maturity, except for **Experimentation** which is essential for the **Exploratory** AI level of maturity. In addition, the following capability areas are also closely related to Workflow Re-Engineering.

- Refer to **Risk and Governance: Risk Management** for goals and practices related to overall risk management around additional security and workflow risks.
- Refer to **Data: Data Lifecycle Management** and **Data Quality Assurance** for goals and practices related to structured and unstructured data to support workflows, especially when agentic approaches are used.
- Refer to **Engineering: AI Architecting, Test and Evaluation**, and **Transparency and Explainability** related to goals and practices for designing and implementing the workflows with sound engineering principles.
- Refer to the **Technology Ecosystem: Supply Chain Management** for goals and practices related to managing and maintaining existing third-party dependencies.

Figure 7 presents an example assessment outcome for the Workflow Re-Engineering dimension, showing the result of each capability area evaluated against maturity levels based on the extent to which the organizational unit meets the relevant goals, practices, and maturity indicators.

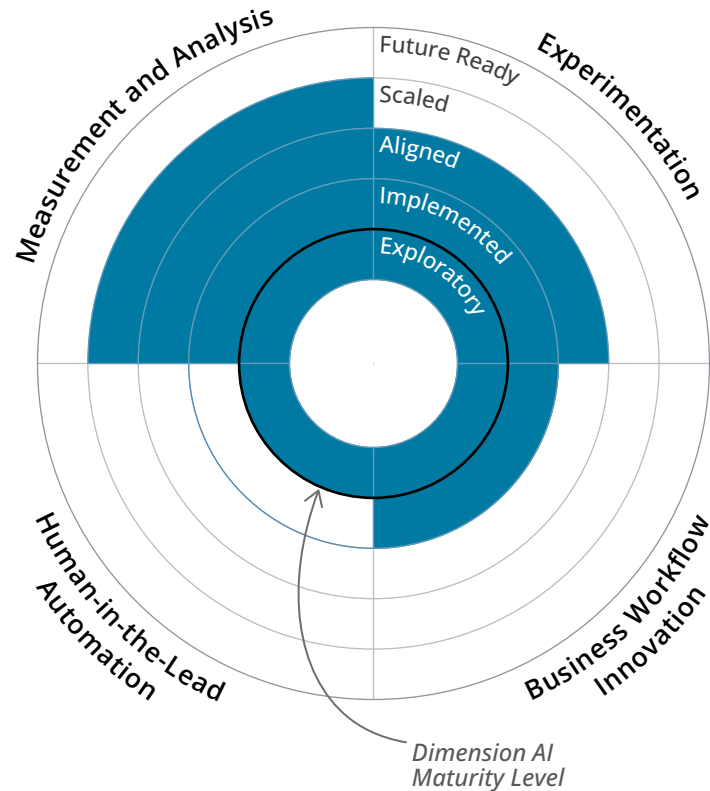


Figure 7: Example Assessment Outcome for the Workflow Re-Engineering Dimension and Capability Areas

4.7 Risk and Governance Dimension

The Risk and Governance dimension capability areas ensure that risks related to AI adoption are managed, AI adoption is executed with appropriate security practices, the relevant policies and standards are followed, and that the organization complies with regulatory obligations that are relevant to AI.

4.7.1 Risk Management Capability Area

*The ability of the organization to identify, understand, and disposition AI-related risks in accordance with the organizational unit’s culture, requirements, and risk tolerance*

Goals and Practices

|        |                                                                                                                                         |                                                                                                                                                                                                           |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | Risks due to the organizational unit's AI adoption are identified, evaluated, and communicated. Processes and practices are in place to |                                                                                                                                                                                                           |
|        | Practice 1                                                                                                                              | Identify types or categories of risks that will be impacted by the organizational unit's AI adoption (e.g., operational, compliance, reputational, third-party, information security, and privacy risks). |
|        | Practice 2                                                                                                                              | Evaluate existing risks which may become more significant or potentially constrain AI adoption efforts.                                                                                                   |
|        | Practice 3                                                                                                                              | Adjust the organization's risk tolerance and appetite in impacted domains, if appropriate.                                                                                                                |
|        | Practice 4                                                                                                                              | Define and communicate to stakeholders the organization's risk appetite or acceptable level of risk for each type or category, including risks inherent to AI adoption.                                   |
|        | Practice 5                                                                                                                              | Make appropriate changes to the management of risks in the impacted domains.                                                                                                                              |
| Goal 2 | An AI risk management process is established. Processes and practices are in place to                                                   |                                                                                                                                                                                                           |
|        | Practice 1                                                                                                                              | Integrate management of AI risks into existing risk processes and systems (e.g., existing risk frameworks and enterprise governance, risk, and compliance [EGRC] solutions).                              |
|        | Practice 2                                                                                                                              | Define and communicate expected controls for AI-specific areas, based on inherent risks (e.g., explainability, human-in-the-lead automation, prompt filtering, bias, or identity).                        |

|        |                                                                                                                            |                                                                                                                                                              |
|--------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 3 | Practice 3                                                                                                                 | Identify AI-related risks as part of routine operational activities.                                                                                         |
|        | Practice 4                                                                                                                 | Assess AI related risks and findings.                                                                                                                        |
|        | Practice 5                                                                                                                 | Disposition AI risks (e.g., mitigate, accept, transfer, monitor).                                                                                            |
|        | Independently test the effectiveness of expected controls relevant to AI adoption. Processes and practices are in place to |                                                                                                                                                              |
|        | Practice 1                                                                                                                 | Identify the expected controls to mitigate risks related to AI adoption, in accordance with the organizational unit's tolerance or risk management guidance. |
|        | Practice 2                                                                                                                 | Adjust the organization's risk tolerance and appetite in impacted domains, if appropriate.                                                                   |
|        | Practice 3                                                                                                                 | Test the design and effectiveness of expected controls.                                                                                                      |
|        | Practice 4                                                                                                                 | Use testing results to identify risks, correct control deficiencies, or identify new controls, as needed.                                                    |

 All the goals and applicable practices of the Risk Management capability area need to be accomplished for this dimension to be at **Implemented AI Level Maturity (Level 2)**.

Example Artifacts

- |                                                                                                                                                                                                                           |                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• risk score</li><li>• hazard and harms identification methods</li><li>• risk assessment approaches (e.g., root cause analysis, System Theoretic Process Analysis [STPA])</li></ul> | <ul style="list-style-type: none"><li>• audit reports of third-party AI partners</li><li>• EGRC strategy</li><li>• risk appetite documentation</li><li>• partners risk appetite documentation</li><li>• AI risk categories</li></ul> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

4.7.2 Policy and Compliance Capability Area

*The ability of the organizational unit to establish and follow relevant policies, adopt AI-related standards relevant to its domain and risk appetite, and adhere to compliance obligations*

Goals and Practices

|               |                                                                                                                                                     |                                                                                                                                                                            |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 1</b> | The organizational unit's policies are aligned to AI adoption. Processes and practices are in place to                                              |                                                                                                                                                                            |
|               | <b>Practice 1</b>                                                                                                                                   | Identify existing organizational policies across domains (e.g., technology, human resources [HR], third-party risk or vendor management) that are relevant to AI adoption. |
|               | <b>Practice 2</b>                                                                                                                                   | Evaluate relevant policies for needed changes based on AI adoption priorities or strategies.                                                                               |
|               | <b>Practice 3</b>                                                                                                                                   | Implement changes and establish new policies as indicated by the evaluation.                                                                                               |
|               | <b>Practice 4</b>                                                                                                                                   | Establish evaluation of AI risks and priorities as part of formal, periodic policy review activities.                                                                      |
| <b>Goal 2</b> | Compliance to relevant industry standards, regulatory requirements, and contractual obligations is managed. Processes and practices are in place to |                                                                                                                                                                            |
|               | <b>Practice 1</b>                                                                                                                                   | Identify the impact of the organization's AI adoption plans on compliance with existing regulatory requirements and industry standards.                                    |
|               | <b>Practice 2</b>                                                                                                                                   | Identify new regulatory requirements or applicable industry standards relating to AI adoption.                                                                             |
|               | <b>Practice 3</b>                                                                                                                                   | Update existing processes, practices, and controls to address regulatory requirements and standards related to AI adoption.                                                |
|               | <b>Practice 4</b>                                                                                                                                   | Monitor and manage AI-related compliance.                                                                                                                                  |



All the goals and applicable practices of the Risk Management capability area need to be accomplished for this dimension to be at **Implemented AI Level Maturity (Level 2)**.

Example Artifacts

- legal playbook
- licenses and ownership of data and AI models
- local, state, federal, and international AI laws
- AI partner agreements and contracts
- organizational policies (e.g., privacy policy, HR policy)
- sector-related AI policies (e.g., automotive, health care)
- policy change log

4.7.3 Responsible AI Capability Area

*The ability to define and enforce the responsible development and use of AI consistently with the organizational unit's mission, dependence on technology, stakeholders, and risk appetite*

Goals and Practices

|               |                                                                                                                  |                                                                                                                                                                                                                                                          |
|---------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 1</b> | Stakeholders that will be impacted by the adoption of AI are identified. Processes and practices are in place to |                                                                                                                                                                                                                                                          |
|               | <b>Practice 1</b>                                                                                                | Identify the populations of users, stakeholders, citizens, and systems that may be affected by the organizational unit's adoption of AI.                                                                                                                 |
|               | <b>Practice 2</b>                                                                                                | Identify any requirements specific to affected populations (e.g., based on legal jurisdiction or the impact to historically vulnerable populations).                                                                                                     |
|               | <b>Practice 3</b>                                                                                                | Consider potential impacts to resource utilization (e.g., energy use and sustainability) that may result from AI adoption.                                                                                                                               |
|               | <b>Practice 4</b>                                                                                                | Evaluate the relevant societal or legal norms, mores, and rules that should be considered.                                                                                                                                                               |
| <b>Goal 2</b> | The organizational unit's approach to responsible AI is defined. Processes and practices are in place to         |                                                                                                                                                                                                                                                          |
|               | <b>Practice 1</b>                                                                                                | Identify the services and products that are relevant to stakeholders and that will be impacted by AI adoption.                                                                                                                                           |
|               | <b>Practice 2</b>                                                                                                | Identify the risks to stakeholders that are inherent to the products and services.                                                                                                                                                                       |
|               | <b>Practice 3</b>                                                                                                | Establish the principles or elements of responsible AI that will be the focus of AI adoption for the assessed organizational unit (e.g., privacy, transparency, bias, fair treatment) for the identified user, stakeholder, citizen groups, and systems. |



|                   |                                                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 3</b>     | The organizational unit's approach to responsible AI is implemented. Processes and practices are in place to                                    |
| <b>Practice 1</b> | Communicate the responsible AI approach to relevant internal and external stakeholders.                                                         |
| <b>Practice 2</b> | Identify the policies, processes, and controls relevant to enforcing the organizational unit's responsible AI approach.                         |
| <b>Practice 3</b> | Change or establish relevant policies, processes, and controls across the development and operation of AI solutions.                            |
| <b>Practice 4</b> | As technology and its use cases evolve, review the responsible AI approach to maintain and communicate alignment with evolving needs and risks. |

 All goals and applicable practices of the Risk Management capability area need to be accomplished for this dimension to be at **Implemented AI Level Maturity (Level 2)**.

**Example Artifacts**

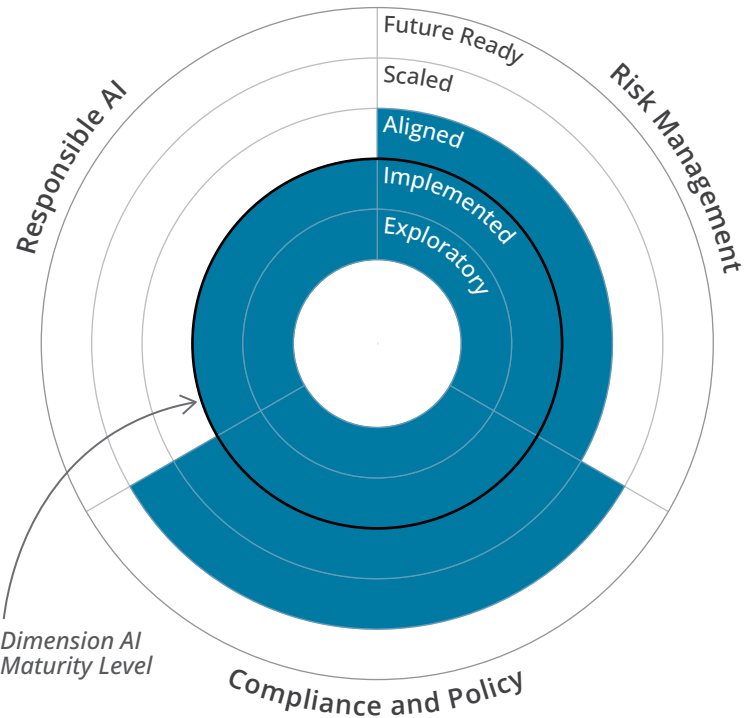
- responsible AI use guidelines
- responsible AI training
- hazard and harms identification methods
- responsible AI design and implementation guidelines
- organizational responsible AI risk policy

**4.7.4 Capability Areas Supporting the Risk and Governance Dimension**

Given the increased categories of risks and uncertainty that AI introduces, all the capability areas of the Risk and Governance dimension are expected to be institutionalized at the **Implemented AI** level of maturity, even when the AI adoption scope is limited. In addition, the following capability areas are closely related and support Risk and Governance.

- Refer to [Engineering: Test and Evaluation](#) for goals and practices related to testing systems for bias.
- Refer to [Operations: AI Model and Agent Security](#) for goals and practices related to securing access to data, resources, and functionalities.

[Figure 8](#) presents an example assessment outcome for the Risk and Governance dimension, showing the result of each capability area evaluated against maturity levels based on the extent to which the organizational unit meets the relevant goals, practices, and maturity indicators.



**Figure 8:** Example Assessment Outcome for the Risk and Governance Dimension and Capability Areas

4.8 Data Dimension

The Data dimension focuses on maintaining relevant, high-quality, and secure structured and unstructured data to ensure data confidentiality, integrity, and availability. The capability areas of this dimension provide timely and relevant data for experimentation, development, training, test, and evaluation to strengthen the value and alignment of AI initiatives

4.8.1 Data Lifecycle Management Capability Area

*The ability to acquire data that are fit for the intended purpose and relevant to the use cases and workflows*

Goals and Practices

|        |                                                                                                                                      |                                                                                                                                                                      |
|--------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | Relevant data are acquired and the provenance of all data are maintained. Processes and practices are in place to                    |                                                                                                                                                                      |
|        | Practice 1                                                                                                                           | Identify, collect, acquire, and/or create data that support development, operation, and testing of AI solutions that support business workflows.                     |
|        | Practice 2                                                                                                                           | Identify and document the sources and categories of data (e.g., type of product, business line) that may be ingested as part of AI initiatives.                      |
|        | Practice 3                                                                                                                           | Document and track datasets and the AI solutions connected to them across all AI initiatives to maintain contextual provenance for structured and unstructured data. |
|        | Practice 4                                                                                                                           | Maintain dataset versioning and document all data transformations, processing, retention, and disposal steps.                                                        |
|        | Practice 5                                                                                                                           | Engage subject matter experts to evaluate data sources and to confirm data relevancy and appropriateness for the intended use case.                                  |
| Goal 2 | Practice 6                                                                                                                           | Develop, maintain, and provide access to up-to-date dataset descriptions, data cards, data dictionaries, metadata, and ontologies as appropriate.                    |
|        | Data stewardship is assigned to formalize the responsibility for the data and its lifecycle. Processes and practices are in place to |                                                                                                                                                                      |
|        | Practice 1                                                                                                                           | Document and mark all systems and data sources that provide authoritative records of data.                                                                           |

|            |                                                                                                                                                                                                                                                             |  |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Practice 2 | Assign owners and subject matter experts to the sources or categories of data.                                                                                                                                                                              |  |
|            | Practice 3                                                                                                                                                                                                                                                  |  |
| Practice 3 | Specify responsibilities for data owners and subject matter experts.                                                                                                                                                                                        |  |
|            | Goal 3                                                                                                                                                                                                                                                      |  |
| Goal 3     | Data privacy and security are maintained throughout the data lifecycle. Processes and practices are in place to                                                                                                                                             |  |
|            | Practice 1                                                                                                                                                                                                                                                  |  |
| Practice 1 | Identify and classify data, including information about people or groups of people, personally identifiable information (PII), personal health information, sensitive company data, and other sensitive data according to applicable regulations and norms. |  |
|            | Practice 2                                                                                                                                                                                                                                                  |  |
| Practice 2 | Restrict access to data based on sensitivity, need-to-know, and least privilege principles.                                                                                                                                                                 |  |
|            | Practice 3                                                                                                                                                                                                                                                  |  |
| Practice 3 | Update and maintain incident response and mitigation plans for data leaks from AI models, agentic systems, or workflows.                                                                                                                                    |  |
|            | Practice 4                                                                                                                                                                                                                                                  |  |
| Practice 4 | Protect data with controls that are commensurate to the organization's threats and AI architectures (e.g., increased reliance on application programming interfaces [APIs] or specific vendors).                                                            |  |
|            | Practice 5                                                                                                                                                                                                                                                  |  |
| Practice 5 | Maintain controls to ensure compliance to data subject rights according to applicable law (e.g., rights to restrict use, right to erasure).                                                                                                                 |  |
|            | Practice 6                                                                                                                                                                                                                                                  |  |
| Practice 6 | Identify and control privacy risks related to data aggregation and analysis by AI solutions.                                                                                                                                                                |  |

 All the goals and applicable practices of the Data Lifecycle Management capability area need to be accomplished for this dimension to be at **Implemented AI Level Maturity (Level 2)**.

Example Artifacts

- data curation guidance
- data cards
- data schema
- dataset description
- data ontology
- data lake criteria
- copyright guidance
- dataset classification guidance
- data usage guidance
- documents marked with classification
- data responsibilities
- data retention policy
- metadata

4.8.2 Data Quality Assurance Capability Area

The ability to ensure that data supporting all AI related activities within the organizational unit are (and remain) fit for the intended purpose, organized, reliable, and usable throughout the entire lifecycle of AI initiatives including ML model development and generative and agentic AI solutions

Goals and Practices

|            |                                                                                                                                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1     | Requirements for data quality are identified. Processes and practices are in place to                                                                                                                                     |
| Practice 1 | Engage all relevant stakeholders to define data quality needs and set expectations for data content.                                                                                                                      |
| Practice 2 | Evaluate data for issues that may result in unwanted outcomes or potential harms.                                                                                                                                         |
| Practice 3 | Evaluate data for inherent bias and document any significant considerations that relate to the intended use case(s).                                                                                                      |
| Practice 4 | Identify the potential harms that may result from misuse of the data, combinations of data, and misuse of the resulting AI system (e.g., to humans, environments, technology) and determine a disposition as appropriate. |
| Goal 2     | Structured and unstructured data in AI systems are prepared for appropriate use. Processes and practices are in place to                                                                                                  |
| Practice 1 | Confirm data relevancy and appropriateness for the intended use case.                                                                                                                                                     |
| Practice 2 | Key contextual data is provided to the operational system (e.g., prompts, RAG lookup, agentic applications).                                                                                                              |
| Practice 3 | Normalize structured data using common practices and formats and select and store novel data points for future analysis.                                                                                                  |
| Practice 4 | Split data into training, testing, and/or validation sets while retaining variation of data across all sets for traditional ML solutions.                                                                                 |
| Practice 5 | Make unstructured data (e.g., text, images, audio, and other non-tabular data) reliable, usable, and governable for downstream use by using standardized formats, clear labels, and metadata.                             |



All the goals and applicable practices of the Data Quality Assurance capability area need to be accomplished for this dimension to be at **Implemented AI Level Maturity** (Level 2).

Example Artifacts

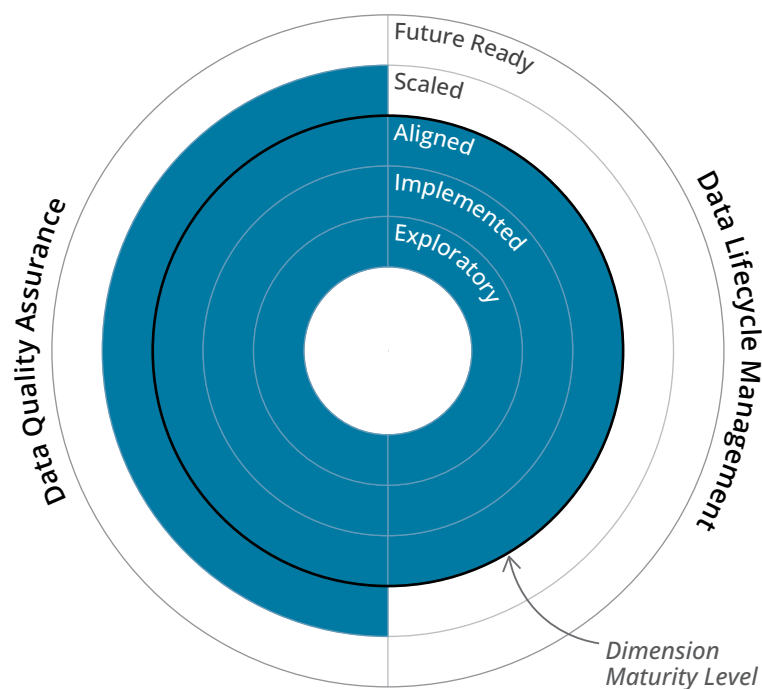
- data labeling guidance
- exploratory data analysis (EDA) outcomes
- extract, transform, load (ETL) pipelines
- bias identification and removal approaches
- metadata

4.8.3 Capability Areas Supporting the Data Dimension

Data is essential to the success of any AI initiative, therefore, the capability areas of the Data dimension all need to be accomplished for **Implemented AI** level of maturity. In addition, the following capability areas support Data related practices.

- Refer to [Risk and Governance: Responsible AI](#) for goals and practices related to use of privacy and biases in data.
- Refer to [Engineering: AI Architecting](#) for goals and practices related to data architecture and data storage.
- Refer to [Engineering: Legacy Integration](#) for goals and practices related to system specific data errors, and [Test and Evaluation](#) for goals and practices related to use of data for test and evaluation.
- Refer to [Operations: Monitoring](#) for goals and practices related to data monitoring and collection from operational settings.

[Figure 9](#) presents an example assessment outcome for the Data dimension, showing the result of each capability area evaluated against maturity levels based on the extent to which the organizational unit meets the relevant goals, practices, and maturity indicators.



**Figure 9:** Example Assessment Outcome for the Data Dimension and Capability Areas

## 4.9 Engineering Dimension

The Engineering dimension focuses on the key capabilities required to transform AI use cases and workflows into deployed systems, including integration and testing of AI components and solutions in production-ready systems.

### 4.9.1 AI Architecting Capability Area

*The ability to develop and integrate AI components and AI-enabled systems to support the business requirements of organizations, ensuring that the resulting product possesses the required quality attributes*

#### Goals and Practices

|                   |                                                                                                                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 1</b>     | Maintainable AI-enabled system architectures which meet stakeholder requirements are developed. Processes and practices are in place to                                                                                                          |
|                   | <b>Practice 1</b> Elicit and document stakeholder requirements for system architectures to support the short- and long-term needs of AI solutions.                                                                                               |
|                   | <b>Practice 2</b> Define the reference architecture of the technology stack to govern the implementation of AI systems and interactions across the technology ecosystems, including those provided by external partners and information systems. |
| <b>Goal 2</b>     | <b>Practice 3</b> Use clearly defined standards and APIs to connect new AI systems to existing ones, using industry-wide standards when possible.                                                                                                |
|                   | Security and operational risks are addressed in systems architecture. Processes and practices are in place to                                                                                                                                    |
|                   | <b>Practice 1</b> Maintain information about expected data flows involving AI-powered systems and agents.                                                                                                                                        |
|                   | <b>Practice 2</b> Include AI-related security specifications in architectural designs (e.g., monitoring, control of data flows, and short-lived or time-bound credentials).                                                                      |
| <b>Practice 3</b> | Define and integrate the approach to AI agent identity, access management, authentication, and authorization to control operational risks.                                                                                                       |
|                   | <b>Practice 4</b> Segment access and apply principles of the separation of duties to AI agents based on their role and identity.                                                                                                                 |



|                   |                                                                                                                                                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 3</b>     | The scalability needs of an AI solution are considered in system design. Processes and practices are in place to                                                                                                                                |
| <b>Practice 1</b> | Define how resources will be vertically or horizontally scaled to support users and AI-related workloads to satisfy scalability expectations.                                                                                                   |
| <b>Practice 2</b> | Determine how resource demand and requirements to support the intended scale of the AI activity will be assessed, supported, and communicated (i.e., sufficient resources to handle data storage, model inference needs, and networking needs). |

 All the goals and applicable practices of the AI Architecting capability area need to be accomplished for this dimension to be at **Implemented AI Level Maturity (Level 2)**.

Example Artifacts

- |                                                          |                                                      |
|----------------------------------------------------------|------------------------------------------------------|
| • architecturally significant requirements               | • internal and external system and data dependencies |
| • prototypes                                             | • shared data sources                                |
| • SLAs                                                   | • version control                                    |
| • reference architecture documentation                   | • logs                                               |
| • technology stack documentation                         | • options and trade off analysis                     |
| • implementation and architecture conformance guidelines | • evolution plan                                     |
|                                                          | • roadmap                                            |
|                                                          | • modernization plan                                 |

4.9.2 Test and Evaluation Capability Area

*The ability to perform verification and validation to provide evidence that the AI solutions meet their technical requirements and support business goals*

Goals and Practices

|                   |                                                                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 1</b>     | AI solution-relevant testing in addition to routine software testing is performed for AI-specific functionalities of systems. Processes and practices are in place to |
| <b>Practice 1</b> | Identify AI-specific functionalities that should be tested.                                                                                                           |
| <b>Practice 2</b> | As needed, test AI functionalities in an appropriately isolated environment.                                                                                          |

|                   |                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Practice 3</b> | Conduct user usability and user acceptance testing (UAT) of AI solutions or make any needed changes to existing UAT.                                 |
| <b>Practice 4</b> | Use regression and integration testing focused on AI capabilities to ensure safe interactions with legacy systems.                                   |
| <b>Practice 5</b> | Ensure that specialized tests are defined to catch AI-specific issues (e.g., data drift and hallucinations).                                         |
| <b>Goal 2</b>     | The ability of the AI solution to perform to the expected quality levels is properly tracked and documented. Processes and practices are in place to |
| <b>Practice 1</b> | Define an appropriate method to baseline and validate outputs of AI functionalities that do not always produce the same result.                      |
| <b>Practice 2</b> | Ensure system and process owners are aware of test results and their level of repeatability.                                                         |
| <b>Practice 3</b> | Ensure that AI-functionality testing can be automated (including the testing of generative AI results).                                              |
| <b>Practice 4</b> | Properly track changes in response to testing outcomes over time and retest as appropriate.                                                          |

 All the goals and applicable practices of the Test and Evaluation capability area need to be accomplished for this dimension to be at **Implemented AI Level Maturity (Level 2)**.

Example Artifacts

- |                                   |                   |
|-----------------------------------|-------------------|
| • test cases                      | • quality tests   |
| • test plan                       | • automated tests |
| • testing approaches              | • baselines       |
| • code reviews                    | • failure modes   |
| • production deployment readiness | • error logs      |

4.9.3 Legacy Integration Capability Area

The ability to assemble AI-enabled products and workflows that enhance the capabilities of legacy systems and interact with existing tools, data, and infrastructure

Goals and Practices

|        |                                                                                                                                                                       |                                                                                                                                                                          |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | The legacy system integration of AI components is achieved following sound architecture approaches and interoperability. Processes and practices are in place to      |                                                                                                                                                                          |
|        | Practice 1                                                                                                                                                            | Elicit system-level needs and different quality attributes from all relevant stakeholders when integrating with legacy systems.                                          |
|        | Practice 2                                                                                                                                                            | Evaluate AI-components for integrability with legacy systems, including the applicability of autonomous AI solutions (i.e., agentic AI).                                 |
|        | Practice 3                                                                                                                                                            | Identify legacy system integration points for AI-components being developed or onboarded for integration, to support implementation.                                     |
|        | Practice 4                                                                                                                                                            | Assess gaps against requirements in the technology stack and equipment, including hardware and infrastructure, access to AI-components, and so on.                       |
| Goal 2 | Practice 5                                                                                                                                                            | Design data processing systems to allow gathering of production data for analysis, test, and training.                                                                   |
|        | The legacy system lifecycle is managed on an ongoing basis to enable the seamless integration with AI systems and components. Processes and practices are in place to |                                                                                                                                                                          |
|        | Practice 1                                                                                                                                                            | Update existing reference architectures to include AI components.                                                                                                        |
|        | Practice 2                                                                                                                                                            | Update legacy infrastructure libraries and APIs regularly to stay in sync with the most recent capabilities and security updates.                                        |
|        | Practice 3                                                                                                                                                            | Retire applications, systems, and infrastructure to optimize technology investments, support AI adoption, and minimize end-of-life (EOL) and end-of-support (EOS) risks. |

⚙️⚙️ All the goals and applicable practices of the Legacy Integration capability area need to be accomplished for this dimension to be at **Aligned AI Level Maturity** (Level 3).

Example Artifacts

- AI model requirements
- data lake criteria
- system environment specifications
- system functional and hardware requirements
- sandbox specifications
- service continuity process
- modernization plan
- legacy system software and system architecture documentation

4.9.4 Transparency and Explainability Capability Area

The ability to ensure that the AI systems deliver understandable, use case appropriate explanations of their behavior and outputs to provide an effective trustworthy experience and traceability to understand agentic solution behaviors

Goals and Practices

|        |                                                                                                                                                                    |                                                                                                                                                                                                     |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | Requirements for transparency and explainability are identified for the AI solutions. Processes and practices are in place to                                      |                                                                                                                                                                                                     |
|        | Practice 1                                                                                                                                                         | Elicit, document, and collect from relevant stakeholders (including the engineering teams) explainability and traceability requirements that support operational, compliance, and enterprise goals. |
| Goal 2 | Practice 2                                                                                                                                                         | Elicit, collect, and document explainability and understandability requirements from end users to support the development of trustworthy systems.                                                   |
|        | Requirements for transparency, explainability and understandability are explicitly addressed in the AI solutions' designs. Processes and practices are in place to |                                                                                                                                                                                                     |
|        | Practice 1                                                                                                                                                         | Follow clearly defined protocols on AI-components to expose the traceability and explainability of AI-specific decisions.                                                                           |
|        | Practice 2                                                                                                                                                         | Establish mechanisms to track and log explainability-related attributes.                                                                                                                            |
|        | Practice 3                                                                                                                                                         | Design component interactions to expose transparency and explainability attributes.                                                                                                                 |

|            |                                                                                                                                                                      |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1     | Explanations provided by AI solutions are relevant and effective for end users of the systems. Processes and practices are in place to                               |
| Practice 1 | Identify content, interactions, and outputs that may be confusing and determine when, where, and how to provide useful explanations of AI system outputs to users.   |
| Practice 2 | Use appropriate interaction design practices and elicit feedback from end users to validate that explanations of the system's outputs are understandable and useful. |

⚙️ All the goals and applicable practices of the Transparency and Explainability capability area need to be accomplished for this dimension to be at **Aligned AI Level Maturity** (Level 3).

Example Artifacts

- transparency and explainability-related design patterns
- transparency and explainability requirements documentation

- end user needs
- privacy and responsible AI guidelines
- relevant API documentation
- data documentation

4.9.5 Capability Areas Supporting the Engineering Dimension

The Engineering dimension spans the **Implemented AI** and **Aligned AI** levels of maturity through its capability levels. In addition, both the **Operations** and **Technology Ecosystem** dimensions support the maturity of the Engineering dimension.

- Refer to **Operations: Monitoring** for goals and practices related to collecting actionable information and logging.
- Refer to **Technology Ecosystem: Supply Chain Management** for goals and practices related to dependencies of other AI products and vendor solutions, to **Technology Infrastructure** for goals and practices related to the underlying infrastructure needs, and to **Deployment at Scale** for goals and practices related to scaling operational deployments.

Figure 10 presents an example assessment outcome for the Engineering dimension, showing the result of each capability area evaluated against maturity levels based on the extent to which the organizational unit meets the relevant goals, practices, and maturity indicators.

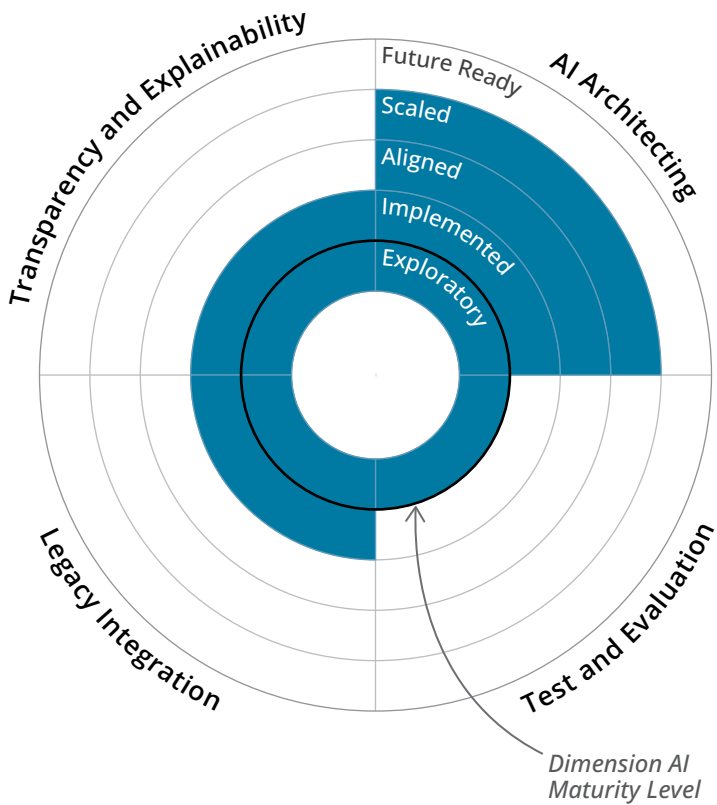


Figure 10: Example Assessment Outcome for the Engineering Dimension and Capability Areas

4.10 Operations Dimension

The Operations dimension brings together capability areas essential for monitoring systems for success, establishing feedback loops that identify future improvements, and managing AI deployments with an emphasis on AI model and agent management and security. These capability areas establish the backbone of effective long-term sustainment and evolution through the collection of relevant, actionable data for continuous improvement.

4.10.1 AI Model Management Capability Area

*The ability to manage the lifecycle of AI models, including their decommissioning*

Goals and Practices

|        |                                                                                                                                                                      |                                                                                                                                                                                                            |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | The version of models used for inference is tracked and controllable at the operation's end point. Processes and practices are in place to                           |                                                                                                                                                                                                            |
|        | Practice 1                                                                                                                                                           | Use version control for the model, related AI, software components, and datasets and track the lineage of data and trained models.                                                                         |
|        | Practice 2                                                                                                                                                           | Archive and make available previous model versions through an API.                                                                                                                                         |
| Goal 2 | Practice 3                                                                                                                                                           | Use a model serving framework that can swap models with minimal to no unintended consequences.                                                                                                             |
|        | Changes to model availability are documented and communicated clearly to relevant stakeholders. Processes and practices are in place to                              |                                                                                                                                                                                                            |
|        | Practice 1                                                                                                                                                           | Have a stable retraining processes that can be run by authorized users.                                                                                                                                    |
| Goal 3 | Practice 2                                                                                                                                                           | Define specific conditions to automatically retrigger retraining based on monitored values.                                                                                                                |
|        | Practice 3                                                                                                                                                           | Archive generative AI model artifacts and metadata (e.g., version, release date, configuration parameters, temperature, seeds, prompts) to ensure consistent performance when generative AI models evolve. |
|        | Gated evaluation conditions are met before models are made available for AI initiatives or are used as part of AI solutions. Processes and practices are in place to |                                                                                                                                                                                                            |
| Goal 1 | Practice 1                                                                                                                                                           | Run an AI initiative defined model test suite as part of the model release process.                                                                                                                        |

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| Practice 2 | Define and implement AI solution tests to detect changes to AI solutions. |
| Practice 3 | Assess and disposition any risks of the AI model deployment or change.    |

 All the goals and applicable practices of the AI Model Management capability area need to be accomplished for this dimension to be at **Implemented AI Level Maturity** (Level 2).

Example Artifacts

- repository describing available AI models (including ML models, generative AI models, and vendor solutions models as applicable)
- model responsibilities
- model usage guidance
- licenses and ownership guidelines of models
- model cards
- privacy and use policies
- model descriptions
- AI model requirements

4.10.2 AI Model and Agent Security Capability Area

*The ability to secure AI solutions, agents, and supporting data to mitigate operational risks, prevent compromises to the integrity of AI solutions and data, and control the actions of AI agents*

Goals and Practices

|        |                                                                                                                                                            |                                                                                                                                                                                                                                                                                                    |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | AI models and agents are protected to prevent compromises or impacts to the integrity of code, AI models, or data. Processes and practices are in place to |                                                                                                                                                                                                                                                                                                    |
|        | Practice 1                                                                                                                                                 | Implement integrity controls (e.g., code signing, code provenance) in AI solution pipelines.                                                                                                                                                                                                       |
|        | Practice 2                                                                                                                                                 | Apply heightened scrutiny for access to models and data (for example applying privileged identity management, multifactor authentication, and appropriate access control models [e.g., attribute and role-based access controls] to entitlements affecting pipelines and production environments). |
| Goal 2 | Practice 3                                                                                                                                                 | Ensure the integrity and security controls for the technology used to support AI solutions, including applicable cybersecurity practices.                                                                                                                                                          |



|                   |                                                                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 2</b>     | Actions by AI models and agents are monitored and controlled. Processes and practices are in place to                               |
| <b>Practice 1</b> | Identify and define the allowable entitlements for AI models, agents, and systems.                                                  |
| <b>Practice 2</b> | Apply identity and access management principles and practices to non-human entities (e.g., system and service accounts, AI agents). |
| <b>Practice 3</b> | Define and implement monitoring and appropriate analytics for the actions of agents as part of AI-solution monitoring.              |



All the goals and applicable practices of the AI Model and Agent Security capability area need to be accomplished for this dimension to be at **Implemented AI Level Maturity** (Level 2).

Example Artifacts

- identity access management guidelines
- AI model and agent repositories
- agent interaction logs

4.10.3 Monitoring Capability Area

*The ability to observe the AI-specific behavior of deployed systems and to collect data about their performance and execution*

Goals and Practices

|                   |                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 1</b>     | Actionable information is collected and used concerning solutions in operation. Processes and practices are in place to |
| <b>Practice 1</b> | Establish monitoring requirements for AI solutions (e.g., performance, drift detection, degradation, and usage).        |
| <b>Practice 2</b> | Capture user feedback on product performance and model functionality.                                                   |
| <b>Practice 3</b> | Collect and store monitoring output for system and human review.                                                        |
| <b>Practice 4</b> | Use monitoring results to improve AI solutions.                                                                         |
| <b>Practice 5</b> | Implement continuous monitoring for agentic and autonomous solutions.                                                   |

|                   |                                                                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 2</b>     | Failures in AI solutions are detected and dispositioned. Processes and practices are in place to                                                              |
| <b>Practice 1</b> | Establish an incident response plan providing guidelines for detection, correlation, escalation, and response to AI solution events, failures, and incidents. |
| <b>Practice 2</b> | Generate automatic failure alerts for recurring data pipelines (automated pipelines running at regular, scheduled intervals).                                 |
| <b>Practice 3</b> | Document roles, responsibilities, and communication channels for responses to AI-related failures.                                                            |
| <b>Goal 3</b>     | Operational data for legal, compliance, accounting, and auditing uses is collected. Processes and practices are in place to                                   |
| <b>Practice 1</b> | Monitor the use of AI systems to detect and identify inappropriate use.                                                                                       |
| <b>Practice 2</b> | Collect evidence of compliance with regulation.                                                                                                               |
| <b>Practice 3</b> | Collect cost and revenue data along with model outputs to support ROI analysis.                                                                               |



All the goals and applicable practices of the Monitoring capability area need to be accomplished for this dimension to be at **Aligned AI Level Maturity** (Level 3).

Example Artifacts

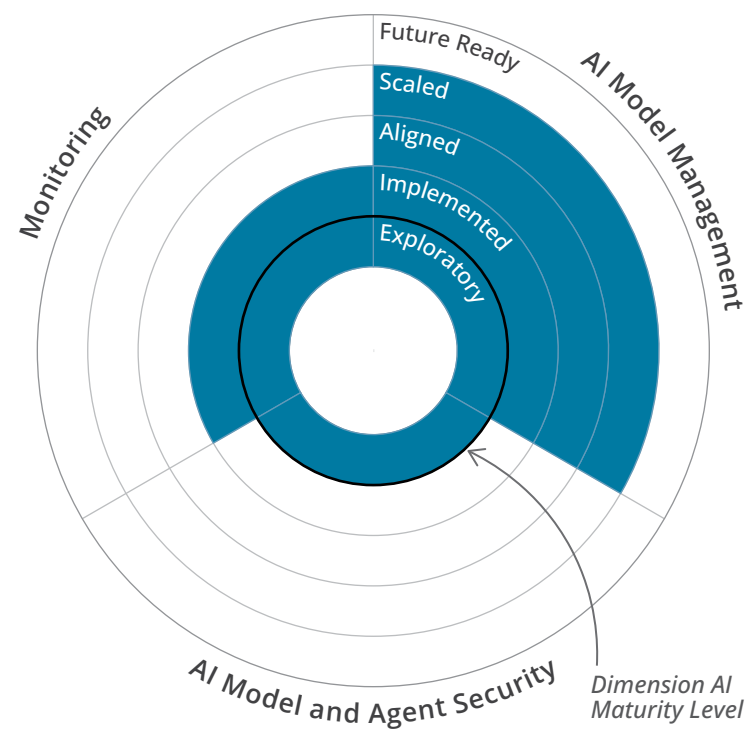
- service level agreements
- monitoring tools and implementations
- monitoring logs
- alerts descriptions

4.10.4 Capability Areas Supporting the Operations Dimension

[AI Model Management](#), [AI Model and Agent Security](#), and [Monitoring](#) are capability areas that are essential for any AI solution, and today any savvy organization which relies on software needs to have these as fundamental capabilities. Therefore, with the exception of Monitoring, they are expected at the **Implemented AI** level of maturity. Monitoring, is deferred to the **Aligned AI** level of maturity to give organizations an opportunity to determine which signals and decisions warrant monitoring to avoid supporting investments that may not pan out as helpful.

- Other capability areas that support Operations include the following.
- Refer to [Risk and Governance: Risk Management](#) for goals and practices related to identification of risks.
  - Refer to [Data: Data Lifecycle Management](#) and [Data Quality Assurance](#) for goals and practices related to acquiring high quality data from operational environments.
  - Refer to [Technology Ecosystem: Supply Chain Management](#) for goals and practices related to third-party models acquired and maintained through supplier agreements.

[Figure 11](#) presents an example assessment outcome for the Operations dimension, showing the result of each capability area evaluated against maturity levels based on the extent to which the organizational unit meets the relevant goals, practices, and maturity indicators.



**Figure 11:** Example Assessment Outcome for the Operations Dimension and Capability Areas

### 4.11 Technology Ecosystem Dimension

The Technology Ecosystem dimension focuses on the organizational unit’s capabilities to manage the AI-relevant dependencies, technologies, and required infrastructure to ensure the efficiency and scalability of AI solutions.

#### 4.11.1 Technology Infrastructure Capability Area

*The ability to provide technology infrastructure supporting the long-term success of AI initiatives*

##### Goals and Practices

|        |                                                                                                                                                                                                           |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal 1 | Technology infrastructure requirements to support AI initiatives are defined. Processes and practices are in place to                                                                                     |
|        | <b>Practice 1</b> Elicit and document infrastructure requirements and computing needs to support AI adoption (e.g., fast storage, high bandwidth, and accelerators).                                      |
|        | <b>Practice 2</b> Assess infrastructure components for trade-offs related to key AI-relevant service level concerns (e.g., resource overload, availability, security, intent traceability, data control). |
| Goal 2 | Periodically reassess and update infrastructure requirements as AI solutions and AI-relevant technologies evolve.                                                                                         |
|        | <b>Practice 1</b> Plan and allocate technology infrastructure.                                                                                                                                            |
|        | <b>Practice 2</b> Make infrastructure resources available for teams to test and implement AI solutions.                                                                                                   |
| Goal 3 | Measure and report infrastructure performance and related costs analysis, including usage, maintenance, service, token limits, and so on.                                                                 |
|        | <b>Practice 1</b> A proactive technology refresh strategy is established. Processes and practices are in place to                                                                                         |
|        | Collect and analyze data on AI technologies and supporting infrastructure to inform technology lifecycle management and refresh decisions.                                                                |

**Practice 2** Define clear timelines to proactively update AI solutions and AI-relevant technologies.

**Practice 3** Upgrade technology, including libraries, AI solutions, APIs, infrastructure, and software, including resolving vulnerabilities and other cybersecurity risks.



All the goals and applicable practices of the Technology Infrastructure capability area need to be accomplished for this dimension to be at **Implemented AI Level Maturity (Level 2)**.

#### Example Artifacts

- deployment views of architecture descriptions
- AI strategy mapping for technology requirements
- capacity plans

### 4.11.2 Supply Chain Management Capability Area

*The ability to identify, analyze, and manage existing supply chain and third-party dependencies related to AI systems and products*

#### Goals and Practices

|               |                                                                                                                                                                      |                                                                                                                                                                                          |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goal 1</b> | An inventory of third-party vendors and dependencies is maintained that includes AI-related third parties. Processes and practices are in place to                   |                                                                                                                                                                                          |
|               | <b>Practice 1</b>                                                                                                                                                    | Maintain an inventory of third parties providing AI-relevant data, models, solutions, and tooling, whether on-boarded to the organization's technology environment or hosted externally. |
|               | <b>Practice 2</b>                                                                                                                                                    | Prioritize third parties by their criticality and impact on AI adoption goals and risk appetite.                                                                                         |
|               | <b>Practice 3</b>                                                                                                                                                    | Track information relevant to AI solutions as part of the inventory (e.g., vendor's use of foundation models, data practices, data hosting location, and fourth or nth parties).         |
|               | <b>Practice 4</b>                                                                                                                                                    | Periodically update the list of third parties and dependencies.                                                                                                                          |
| <b>Goal 2</b> | Assess third parties and suppliers to ensure compliance with organizational requirements and risk appetite for AI solutions. Processes and practices are in place to |                                                                                                                                                                                          |
|               | <b>Practice 1</b>                                                                                                                                                    | Identify and track requirements for third parties (e.g., data sharing, security, responsible AI, human-centeredness, or model fitness).                                                  |

**Practice 2** Monitor or assess third-party performance against the organization's requirements and contracts based on prioritization.

**Practice 3** Establish workarounds and termination plans in the event of third-party failures based on a risk and criticality prioritization.

**Goal 3** External data dependencies are managed. Processes and practices are in place to

- |                   |                                                                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Practice 1</b> | Identify and track data used as part of AI solutions that originates outside of internal data management practices, including external entities and non-traditional data sources. |
| <b>Practice 2</b> | Integrate external data sources into the organization's management of third parties.                                                                                              |
| <b>Practice 3</b> | Track and apply data management practices to data collected as a part of using third-party services.                                                                              |



All the goals and applicable practices of the Supply Chain Management capability area need to be accomplished for this dimension to be at **Aligned AI Level Maturity (Level 3)**.

#### Example Artifacts

- demand and capacity planning documents
- standard operating procedures
- software bill of materials (SBOMs)
- compliance checklists
- AI model lineage
- dependency registries
- supplier API specifications
- transaction logs
- traceability and provenance records

### 4.11.3 Deployment at Scale Capability Area

*The ability to deploy and scale technology and AI solutions across the organizational unit to support strategy and AI-native workflows, while maximizing ROI and minimizing technology change risks*

#### Goals and Practices

|               |                                                                                                                              |                                                                                               |
|---------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Goal 1</b> | Scaling AI solutions to support the organization is planned and properly documented. Processes and practices are in place to |                                                                                               |
|               | <b>Practice 1</b>                                                                                                            | Identify AI solutions that add value across the organization and plan for scaled deployments. |

|                   |                                                                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Practice 2</b> | Identify and manage the risks of organizational and technology changes due to scaling, including costs and performance.                                                        |
| <b>Practice 3</b> | Inventory the existing solutions to support de-duplication through rationalization, transition, and decommissioning of the duplicative solutions to support scaled deployment. |
| <b>Practice 4</b> | Identify the required technology changes needed to satisfy the solutions planned to scale.                                                                                     |
| <b>Practice 5</b> | Evaluate the progress of current AI solution deployments, including operations and maintenance in scaling.                                                                     |
| <b>Goal 2</b>     | Performance and reliability targets for AI solutions are established and met. Processes and practices are in place to                                                          |
| <b>Practice 1</b> | Establish SLAs and service-level objectives (SLOs) for AI solutions.                                                                                                           |
| <b>Practice 2</b> | Ensure capacity is managed in accordance with the criticality of scaled solutions.                                                                                             |
| <b>Practice 3</b> | Use processes and practices that enable AI solutions to provide continued service under different conditions.                                                                  |
| <b>Practice 4</b> | Use proactive notifications to inform stakeholders of system outages.                                                                                                          |
| <b>Practice 5</b> | Update and maintain business continuity plans, project and program deployment, and dependencies to guide seamless operations of scaled solutions.                              |

 All goals and applicable practices of Deployment at Scale capability area need to be accomplished for this dimension to be at **Scaled AI Level Maturity** (Level 4).

**Example Artifacts**

- scalability requirements
  - technology, project management, and resources
- planning processes supporting scaled AI solutions
  - duplicative solution lists

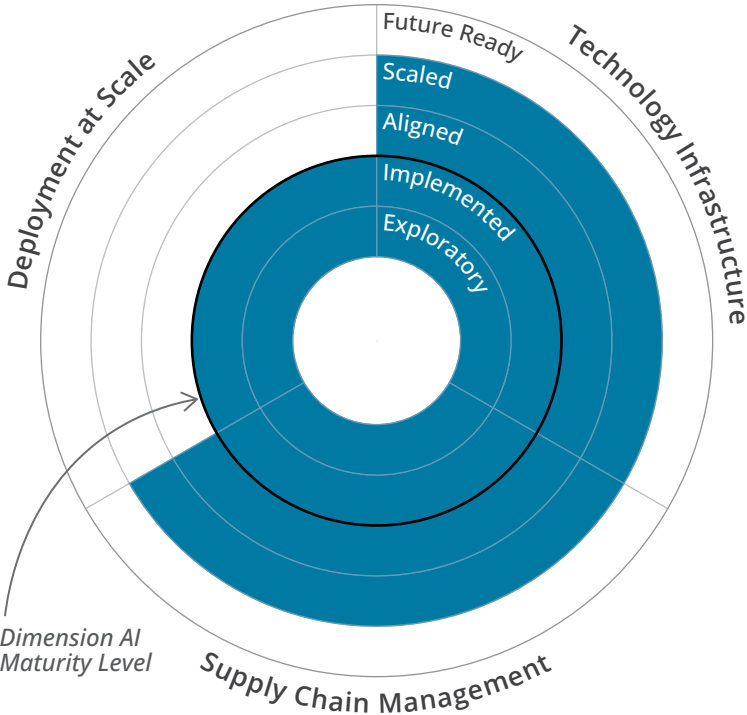
**4.11.4 Capability Areas Supporting the Technology Ecosystem Dimension**

The Technology Ecosystem evolves as the maturity increases and the corresponding needs expand. The capability areas in this dimension reflect that progression by spanning multiple maturity levels from **Implemented AI** to **Scaled AI**.

Other capability areas that support Technology Ecosystem include the following.

- Refer to [Organizational Strategy: AI Partnership Development](#) for goals and practices related to selecting strategic partners in alignment with AI initiatives.
- Refer to [Engineering: AI Architecting](#) for goals and practices related to architectural concerns such as integration and system decomposition.
- Refer to [Operations: Monitoring](#) for goals and practices related to monitoring and feedback collection practices.

[Figure 12](#) presents an example assessment outcome for the Technology Ecosystem dimension, showing the result of each capability area evaluated against maturity levels based on the extent to which the organizational unit meets the relevant goals, practices, and maturity indicators.



**Figure 12:** Example Assessment Outcome for the Technology Ecosystem Dimension and Capability Areas



4.12 Dimensions Summary

Figure 13 presents an example of a randomized consolidated assessment summary of the capability areas illustrated in this section and their associated dimensions.

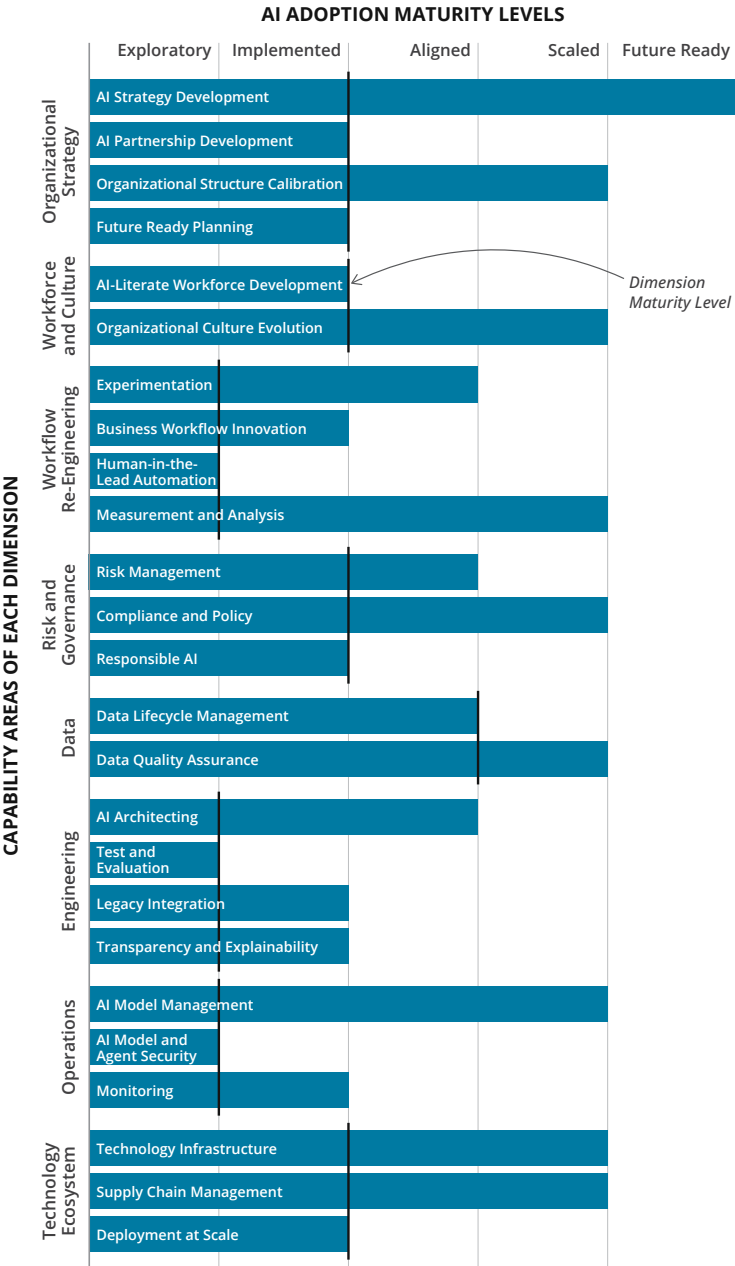


Figure 13: Example Assessment Summary

4.13 Maturity Indicators

Maturity indicators allow collecting verifiable attributes, metrics, or observable characteristics to assess the extent to which a capability area satisfies the criteria associated with a defined maturity level along with the goals and practices pertaining to the capability. Maturity indicators provide the objective evidence of institutionalization, performance, and repeatability, enhancing the reliable evaluation of progression across maturity levels.

The following describes maturity levels, as defined in in Section 3, applied to the perspective of capability areas.

- Exploratory AI:** At this level, the organizational unit is learning about the identified scope, risks, and adoption of AI for this capability area. Some teams can accomplish some area goals, however, a capability at the Exploratory AI level is not widespread or institutionalized in the organizational unit. Capability area implementation may be ad hoc at this level.
- Implemented AI:** An Implemented AI capability area is planned and executed in accordance with plans, has clearly defined roles, and has adequate resources to produce controlled outputs. Practices to support the capability area goals can be performed by multiple relevant teams on one or more projects supported by a process description.
- Aligned AI:** An Aligned AI capability area is executed, managed, and monitored in multiple adoptions of AI. The achievement of project-specific objectives is closely tied to organizational unit goals and strategies, and metrics used are aligned with business outcomes, tracking ROI allowing for measured consistency.
- Scaled AI:** A Scaled AI capability is not only consistently performed throughout an organizational unit but also can be rapidly scaled. These capabilities are based on a standard processes and policies and have organizational tailoring guidelines. Typically, in an organization with multiple sub-components, the leaders track and share information about ROI and performance in the area.
- Future Ready AI:** An organizational unit with the Future Ready AI capability has a proven record of incorporating new technologies, an existing long term strategic plan, and the ability to shape its future operational environment. For this area, there is a focus on planning, resourcing, and governance to avoid strategic surprise and on leveraging AI adoption as a key component of future growth with predictable performance.

The following three maturity indicators are included in the AI Adoption Maturity Model:

1. The **Accountability** maturity indicator focuses on the accountability of the leaders, managers, and staff of the organization to establish and improve processes to achieve the goals in the capability area.
2. The **Planning** maturity indicator focuses on making sure there are artifacts and guidance to provide consistency, assurance, and predictability in how goals in the capability area are being achieved.
3. The **Resourcing** maturity indicator focuses on the organization providing adequate resources to operate the processes that make up the capability area. Resources include staff, funding, and technology.

The following statements summarize what is expected of each maturity level for each of the maturity indicators for the capability areas.

### Accountability

|                                   |                                                                                                                                                                  |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Level 1<br><b>Exploratory AI</b>  | Accountability is not consistently assigned for the capability area.                                                                                             |
| Level 2<br><b>Implemented AI</b>  | Roles are consistently assigned that are responsible for accomplishing the goals of the capability area.                                                         |
| Level 3<br><b>Aligned AI</b>      | Objectives and related metrics (such as KPI, KRI, ROI) are established for the capability area and reported to higher level management.                          |
| Level 4<br><b>Scaled AI</b>       | Capability area process(es) are measured and managed consistently across organizational unit sub-components.                                                     |
| Level 5<br><b>Future Ready AI</b> | Opportunities for improving the standard process(es) are evaluated and managed (selected and implemented) based on the organizational unit's long-term strategy. |

### Planning

|                                   |                                                                                                                                                   |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Level 1<br><b>Exploratory AI</b>  | Planning for the capability area is not consistently performed.                                                                                   |
| Level 2<br><b>Implemented AI</b>  | There is a plan(s) for accomplishing the goals of the capability area that identifies relevant stakeholders.                                      |
| Level 3<br><b>Aligned AI</b>      | Plans are periodically updated based on the performance of the capability area to satisfy objectives and related metrics (such as KPI, KRI, ROI). |
| Level 4<br><b>Scaled AI</b>       | Plans and policies include consistent definitions for process(es) that can be tailored for use by organizational unit sub-components.             |
| Level 5<br><b>Future Ready AI</b> | Planning considers possible future changes or scenarios (e.g., regulatory, societal, technological) that impact the capability area.              |

### Resourcing

|                                   |                                                                                                                                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Level 1<br><b>Exploratory AI</b>  | Resources for the capability area are not consistently allocated.                                                                                     |
| Level 2<br><b>Implemented AI</b>  | Adequate resources are allocated to the capability area by the established budgeting process.                                                         |
| Level 3<br><b>Aligned AI</b>      | Resources are allocated based on the measured performance of the capability.                                                                          |
| Level 4<br><b>Scaled AI</b>       | Resources are consistently allocated and managed across organizational unit sub-components.                                                           |
| Level 5<br><b>Future Ready AI</b> | Resources are allocated to the capability area to implement the organizational unit's long-term strategy to maximize utility across possible futures. |

Higher maturity (e.g., Scaled AI) supports and correlates to the ability to more easily and reliably scale adoption across the entire organizational unit or enterprise. Take, as an example, a multi-national corporation with multiple divisions. The Scaled AI level of maturity means that—for example—the various divisions need not use the same standards or technology to secure AI agents. But to achieve Level 4—or Scaled AI—maturity they must manage and govern this area according to processes that are planned, managed, measured, and supported consistently across the enterprise. The purpose or “point of” scaled maturity is to support agility across the entire enterprise so that systems, resources,

and processes can more quickly, easily, and reliably support AI adoption and business goals across the enterprise. Enterprises with strong processes that are commonly understood are better able to evolve and change quickly without key work being “lost in translation.”

Assessments or uses of the AI Adoption Maturity Model that are focused on smaller organizations or some organizational sub-component tend to satisfy the questions and maturity indicator as presented in this model more easily. This is because with a smaller organizational scope it is simply easier to coordinate work and make it more consistent and standardized.

#### 4.14 Reaching New Levels of Maturity

While all capability areas are essential for sustainable AI adoption, organizational units need guidance for prioritizing investments and improvement efforts to advance in maturity effectively. These priorities are shaped by both business objectives and the expectations associated with each maturity level. Dependencies between capability areas are important because successful AI adoption is not achieved through isolated improvements. Most capabilities rely on other capabilities being established first or evolving in parallel. Ignoring these dependencies can result in organizations investing in advanced capabilities before the foundational practices needed to support them are in place, leading to increased risk, reduced value realization, and slower adoption progress.

For example, an organization may seek to scale AI across multiple business units, but without clear governance, workforce readiness, architecture standards, or measurement practices, scaling efforts are likely to produce inconsistent outcomes. Similarly, implementing AI-enabled workflows without aligning them to the business strategy can create technical success without a corresponding business value.

For this reason, the AI Adoption Maturity Model highlights dependencies across dimensions and capability areas. These dependencies help organizations understand AI adoption as an interconnected system of capabilities rather than a collection of standalone practices, enabling a more coordinated, sustainable, and successful roadmap for maturity progression. (This is outlined further in [Section 5.2.](#)) At the same time, maturity progression follows a practical sequence that enables capabilities to build on one another effectively. [Table 2](#) illustrates the minimum capability areas required at each maturity level, in addition to the expectation that all capability areas satisfy the maturity indicators associated with that level.

Table 2: Minimal Capability Area Expectations to Advance in Maturity

| POSITIONED AT MATURITY LEVEL                                                                                     | DIMENSION               | CAPABILITY AREA                      |
|------------------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------|
|  Level 1<br>Exploratory AI    | Workflow Re-Engineering | Experimentation                      |
|                                                                                                                  | Workforce and Culture   | AI-Literate Workforce Development    |
|  Level 2<br>Implemented AI    | Data                    | Data Lifecycle Management            |
|                                                                                                                  |                         | Data Quality Assurance               |
|                                                                                                                  | Engineering             | AI Architecting                      |
|                                                                                                                  |                         | Test and Evaluation                  |
|                                                                                                                  | Operations              | AI Model Management                  |
|                                                                                                                  |                         | AI Model and Agent Security          |
|                                                                                                                  | Organizational Strategy | AI Strategy Development              |
|                                                                                                                  | Risk and Governance     | Risk Management                      |
|                                                                                                                  |                         | Policy and Compliance                |
|                                                                                                                  |                         | Responsible AI                       |
|  Level 3<br>Aligned AI        | Technology Ecosystem    | Technology Infrastructure            |
|                                                                                                                  | Engineering             | Legacy Integration                   |
|                                                                                                                  |                         | Transparency and Explainability      |
|                                                                                                                  | Operations              | Monitoring                           |
|                                                                                                                  | Organizational Strategy | AI Partnership Development           |
|                                                                                                                  |                         | Organizational Structure Calibration |
|                                                                                                                  | Technology Ecosystem    | Supply Chain Management              |
|                                                                                                                  | Workflow Re-engineering | Business Workflow Innovation         |
|                                                                                                                  |                         | Measurement and Analysis             |
|                                                                                                                  |                         | Human-in-the-Lead Automation         |
|  Level 4<br>Scaled AI       | Workforce and Culture   | Organizational Culture Evolution     |
|                                                                                                                  | Technology Ecosystem    | Deployment at Scale                  |
|  Level 5<br>Future Ready AI | Organizational Strategy | Future Ready Planning                |

# 5 Measurement Framework and Roadmap

## Measurement Framework and Roadmap

The maturity measurement framework used in the AI Adoption Maturity Model reflects the lessons learned from the SEI's decades long assessments and scoring experiences. Different scoring and assessment methodologies can be used with this model, but in the default method, a dimension is rated to its lowest ranking capability areas, and a capability area is rated to its lowest ranking maturity indicator, even if all the goals and practice areas are satisfied. While this is a harsh scoring mechanism, it reflects the care that was given to the capability areas and goals included in this model to focus on core areas that will stand the test of time, even as technologies evolve. Practices may evolve, but goals and capability areas will likely not.

Goal ratings are derived from multiple evidence streams, including normalization of assessor evaluations, cross-validation with organizational unit survey data, and verification through supporting artifacts. This rigor ensures that baselines are accurately established, enabling organizations to prioritize the most critical activities required for achieving repeatable, consistent, trustworthy, and scalable AI practices. These mechanisms ensure that maturity determinations remain consistent, reproducible, objective, and comparable across assessments, and that organizations can reliably reassess themselves over time.

### 5.1 Rethinking Maturity Assessments in the Era of AI

Today, AI is reshaping a wide range of workflows, including how analyses, evaluations, and assessments are conducted. Despite these advancements, expert-led evaluation, analysis, and elicitation remain foundational techniques that have delivered value for decades across disciplines such as software engineering, process improvement, and, more recently, AI adoption. These methods continue to be essential for assessing whether infrastructure is fit for purpose, eliciting and refining requirements, designing user experiences, developing strategy and supporting artifacts, conducting architecture evaluations, and defining and improving processes. Maturity assessments and design reviews are representative applications of these techniques.

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The maturity measurement framework used in the AI Adoption Maturity Model reflects the lessons learned from the SEI's decades long assessments and scoring experiences.



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An effective maturity assessment is composed of several critical elements: a well-defined maturity model that provides structural rigor, a disciplined approach to evidence collection, and an analytical method that integrates quantitative signals with qualitative insights.

The value of expert-driven assessment approaches lies in their ability to

- strengthen stakeholder communication and alignment
- synthesize insights across otherwise fragmented systems, processes, and organizational boundaries
- systematically uncover latent issues, including implicit requirements, inconsistencies, risks, constraints, and operational bottlenecks that are often undocumented or unarticulated
- identify and prioritize actionable improvements based on organizational goals, target states, and observed maturity gaps

An effective maturity assessment is composed of several critical elements: a well-defined maturity model that provides structural rigor, a disciplined approach to evidence collection, and an analytical method that integrates quantitative signals with qualitative insights. Assessment outcomes are derived through the triangulation of multiple data sources, including stakeholder workshops (both small and large group), structured interviews, enterprise surveys, artifact and documentation analysis, metrics evaluation, tooling and telemetry data, and systematic calibration of findings.

While defined assessment methodologies provide consistency, execution is inherently adaptive. Data collection techniques are selected based on assessment scope, organizational context, and available evidence. For example, in some environments, tooling data may partially or fully substitute for traditional document analysis. Targeted assessments may rely on focused workshops, whereas enterprise-scale evaluations require a combination of interviews, workshops, and cross-unit sampling to validate consistency and reduce bias. This multi-source, evidence-based approach is a defining characteristic that distinguishes maturity assessments from static questionnaires. Maturity assessments do not simply capture responses—they identify capability gaps, define required practices and processes, map their interdependencies, and uncover undocumented and unspoken realities.

Advances in automation and AI-enabled tooling create meaningful opportunities to augment assessment activities. These tools can assist with synthesizing large volumes of data, identifying dependency patterns, correlating findings, and generating traceable insights and narratives. However, such tools function strictly as analytical accelerators—not as assessors. Expert judgment remains central to interpreting context, validating findings, and ensuring that conclusions are grounded in organizational realities.

Emerging agentic and AI-assisted tools have a role to enhance the consistency, scalability, and efficiency of assessment execution while reducing manual analytical effort. Their function is not to autonomously ingest organizational artifacts and produce authoritative assessment outcomes which produces countless security and consistency risks. Automated summarization and pattern extraction represent only one component of a broader, expert-led evaluation process that requires critical reasoning, contextual understanding, and domain expertise. Automated tools can improve consistency and reduce analysis time, but they do not replace the human-in-the-lead approach.

## 5.2 Developing the Roadmap

While great care has gone into the measurement framework and assessment levels, the value-added top priority goal of this AI Adoption Maturity Model effort is to create an actionable roadmap to drive institutional AI adoption. A roadmap is the natural outcome of assessing the dimensions for maturity by focusing on key capability areas and their dependencies and priorities. However, a roadmap is not simply fulfilling all the goals and checking the boxes of maturity indicators. Over time, that approach erodes the effectiveness of maturity model-based assessments, shifting them away from their core objective of producing repeatable, consistent practice with value and ROI outcomes and reducing them to exercises in documentation tracking rather than true capability measurement. The capability areas, along with their supporting goals and practices, are deliberately selected and empirically validated to ensure focus on the most critical challenge areas, rather than constructing a redundant framework.

A successful roadmap starts with a clearly defined target state, not just a desire to “increase maturity.” After completing an assessment, the identified gaps provide a structured view of current capabilities versus expected practices, and an initial estimate of the effort required to advance. However, progressing to the next maturity level should not be treated as an automatic or uniform objective.

Instead, the organization must first determine what level of maturity is appropriate given its business goals, risk posture, regulatory environment, and operational context. This requires explicitly mapping capabilities to outcomes—for example, understanding which maturity levels are necessary to support scalability, ensure reliability, or meet security and compliance requirements. In technical terms, this means

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While great care has gone into the measurement framework and assessment levels, the value-added top priority goal of this AI Adoption Maturity Model effort is to create an actionable roadmap to drive institutional AI adoption.

This approach ensures that maturity advancement is intentional, context-aware, and outcome-driven, rather than a generic progression through predefined levels.

defining the target operating model: the set of workflows, controls, data practices, and engineering behaviors that must be in place to achieve the desired performance and risk thresholds.

Importantly, the target maturity level may vary across dimensions. For instance, an organization may require high maturity in data governance and security to mitigate risk, while maintaining a lower, more experimental maturity level in AI model development to preserve innovation speed. This selective targeting avoids over-engineering and ensures that investments are aligned with the areas that most directly impact business outcomes.

Therefore, before prioritizing gap closure, the organization should

- Define dimension-specific target states based on business and technical requirements.
- Establish measurable criteria for those target states (e.g., reproducibility, auditability, latency, cost efficiency).
- Identify dependencies across capabilities (e.g., data quality enabling reliable business workflow innovation).

Only after this target state is clearly articulated can gaps be meaningfully translated into a roadmap. This approach ensures that maturity advancement is intentional, context-aware, and outcome-driven, rather than a generic progression through predefined levels.

Once a target state is defined, a roadmap is generated based on two inputs: (1) an effort versus value mapping of each capability area based on identified assessment gaps, and (2) the capability area dependencies.

The organizational context, dependencies, and maturity levels allow organizations to come up with a roadmap that allows them to focus their attention based on effort versus value—prioritizing transformational capability areas and quick wins with high ROI to those that are good to have with lower ROI. These are organization specific, based on their maturity levels and gaps along with their goals. A sanitized example from the early adopter engagements can be found in [Figure 14](#).

[Table 3](#) summarizes a dependency mapping of the capability area dependencies of the AI Adoption Maturity Model. Dependencies are defined as a linkage relationship across capability areas where some response or measurement may affect the reciprocal capability area.

We see from the early adopter program that these dependencies are leading to initial roadmap patterns. For example, we see that if an organizational unit is not mature in the [AI Strategy Development](#) capability area, most other dimensions in the Organizational Change dimensions ([Section 4.2.1](#)) mirror its maturity. A low maturity in [Risk Management](#) often also reflects as low maturity in [Business Workflow Innovation](#). As the AI Adoption Maturity Model is new, these patterns from practice are only now emerging and our goal is to share further validated patterns of maturity in the near future.

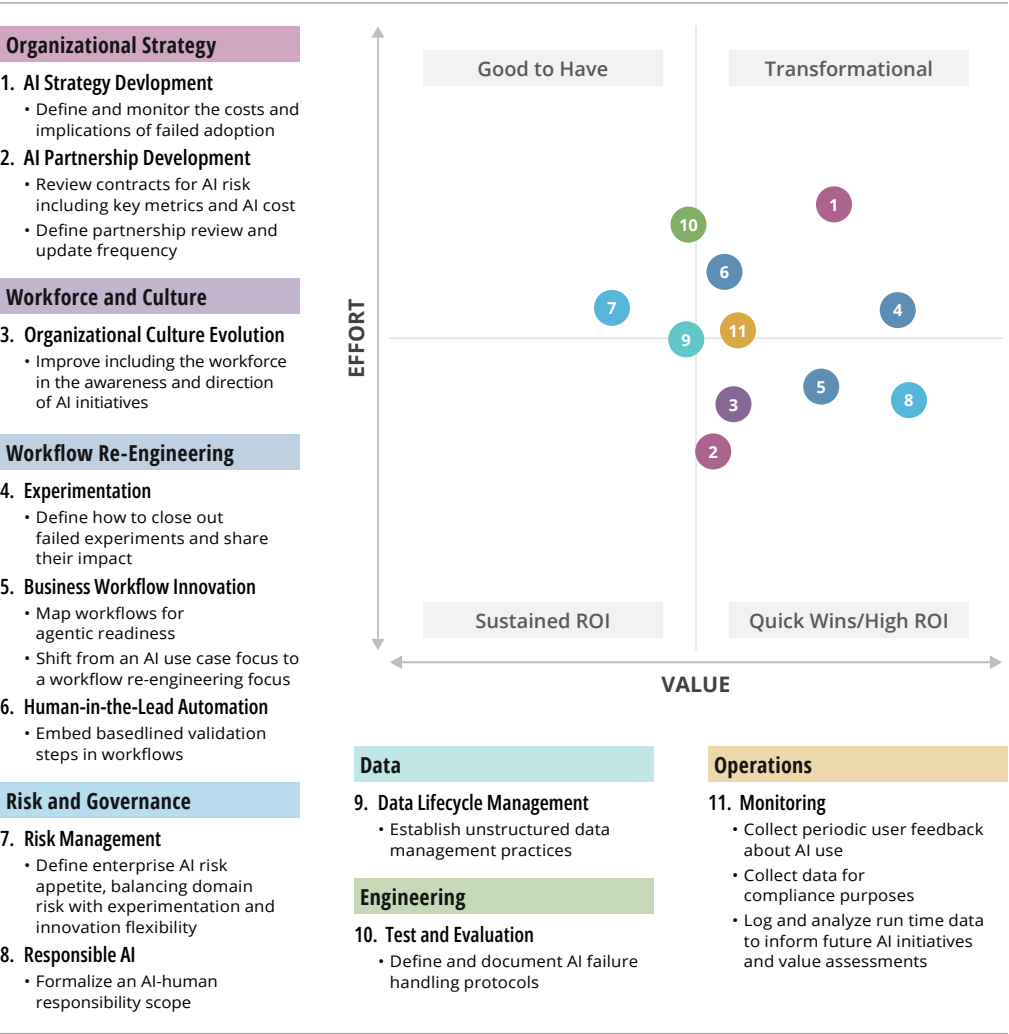


Figure 14: Example Roadmap Showing Capability Area Prioritization

Table 3: Capability Area Dependency Mapping

| ORGANIZATIONAL CHANGE    |                         |                                      |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|--------------------------|-------------------------|--------------------------------------|----------------------------|--------------------------------------|-----------------------|-----------------------------------|----------------------------------|-----------------|-------------------------------|------------------------------|--------------------------|-----------------|-----------------------|
| DIMENSIONS               |                         | ORGANIZATIONAL STRATEGY              |                            |                                      |                       | WORKFORCE & CULTURE               |                                  |                 |                               | WORKFLOW RE-ENGINEERING      |                          |                 |                       |
|                          |                         | AI Strategy Development              | AI Partnership Development | Organizational Structure Calibration | Future Ready Planning | AI-Literate Workforce Development | Organizational Culture Evolution | Experimentation | Business Workflow Engineering | Human-in-the-Lead Automation | Measurement and Analysis | Risk Management | Policy and Compliance |
| ORGANIZATIONAL CHANGE    | ORGANIZATIONAL STRATEGY | AI Strategy Development              |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | AI Partnership Development           |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Organizational Structure Calibration |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Future Ready Planning                |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          | WORKFORCE AND CULTURE   | AI-Literate Workforce Development    |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Organizational Culture Evolution     |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          | WORKFLOW RE-ENGINEERING | Experimentation                      |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Business Workflow Innovation         |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Human-in-the-Lead Automation         |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Measurement and Analysis             |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          | RISK AND GOVERNANCE     | Risk Management                      |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Policy and Compliance                |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Responsible AI                       |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
| AI LIFECYCLE ENGINEERING | DATA                    | Data Lifecycle Management            |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Data Quality Assurance               |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          | ENGINEERING             | AI Architecting                      |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Test and Evaluation                  |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Legacy Integration                   |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Transparency and Explainability      |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          | OPERATIONS              | AI Model Management                  |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | AI Model and Agent Security          |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Monitoring                           |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          | TECHNOLOGY ECOSYSTEM    | Technology Infrastructure            |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Supply Chain Management              |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |
|                          |                         | Deployment at Scale                  |                            |                                      |                       |                                   |                                  |                 |                               |                              |                          |                 |                       |

● Indicates a relationship across capability areas where some responses/measurements may affect the reciprocal capability area

| AI LIFECYCLE ENGINEERING |                         |                                      |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|--------------------------|-------------------------|--------------------------------------|------------------------|---------------------------------|--------------------|---------------------|-----------------|---------------------|-----------------------------|------------|---------------------------|-------------------------|---------------------|
| DIMENSIONS               |                         | DATA                                 |                        |                                 |                    | ENGINEERING         |                 |                     |                             | OPERATIONS |                           |                         |                     |
|                          |                         | Data Lifecycle Management            | Data Quality Assurance | Transparency and Explainability | Legacy Integration | Test and Evaluation | AI Architecting | AI Model Management | AI Model and Agent Security | Monitoring | Technology Infrastructure | Supply Chain Management | Deployment at Scale |
| ORGANIZATIONAL CHANGE    | ORGANIZATIONAL STRATEGY | AI Strategy Development              |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | AI Partnership Development           |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Organizational Structure Calibration |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Future Ready Planning                |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          | WORKFORCE AND CULTURE   | AI-Literate Workforce Development    |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Organizational Culture Evolution     |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          | WORKFLOW RE-ENGINEERING | Experimentation                      |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Business Workflow Innovation         |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Human-in-the-Lead Automation         |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Measurement and Analysis             |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          | RISK AND GOVERNANCE     | Risk Management                      |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Policy and Compliance                |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Responsible AI                       |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
| AI LIFECYCLE ENGINEERING | DATA                    | Data Lifecycle Management            |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Data Quality Assurance               |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          | ENGINEERING             | AI Architecting                      |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Test and Evaluation                  |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Legacy Integration                   |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Transparency and Explainability      |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          | OPERATIONS              | AI Model Management                  |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | AI Model and Agent Security          |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Monitoring                           |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          | TECHNOLOGY ECOSYSTEM    | Technology Infrastructure            |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Supply Chain Management              |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |
|                          |                         | Deployment at Scale                  |                        |                                 |                    |                     |                 |                     |                             |            |                           |                         |                     |

# 6

## Conclusion and Model Evolution

## Conclusion and Model Evolution

The journey to effective AI adoption is complex, and while countless reflections and lessons learned exist, they often lack the actionable clarity organizations need to drive meaningful progress. The AI Adoption Maturity Model seeks to bridge that gap by offering a structured, measurement-based, practical framework that helps organizations pinpoint their most urgent areas for improvement and take informed steps forward. By focusing on challenges, such as establishing baselines, measuring progress, redesigning workflows, and managing the broader AI ecosystem, the model provides intentional guidance rooted in discipline and proven practices rather than quick fixes.

Although many organizations continue grappling with inconsistent results in their AI efforts, the opportunity to capture AI's full potential remains substantial. With deliberate, sustained commitment, organizations can advance their capabilities in predictable and measurable ways.

This first version of the AI Adoption Maturity Model represents an important starting point—but not the final word. As AI technologies, governance requirements, and organizational needs continue to evolve, so too will this model. Ongoing iteration, informed by real-world applications, research insights, and user feedback, will ensure that the model remains relevant, practical, and responsive to the rapidly shifting AI landscape. In addition, our next steps include sharing data derived from applying the AI Adoption Maturity Model, along with key implementation insights, such as capability dependency patterns, common roadmap trajectories, and how a maturity-based approach can inform and optimize partnership strategies. By continually refining and expanding this framework, we aim to support organizations on a long-term, adaptive journey toward responsible, effective, and scalable AI adoption.

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With deliberate, sustained commitment, organizations can advance their capabilities in predictable and measurable ways.



# Glossary

*The Glossary does not include terms that have generally accepted definitions.*

**Agent**—a goal-oriented computing entity that takes actions autonomously to perceive, interact with, and change its environment and create content. Agents can also access and use software tools, recognize patterns, make decisions, and plan tasks.

**Agentic AI**—a system driven by AI agents, that uses a natural language interface and takes actions autonomously (i.e., human prompts or continuous oversight are not necessary) to make decisions and plans multi-stage tasks. agentic AI systems are composed of multiple AI agents and have access to multiple tools.

**Analytic methods**—processes or techniques applied to an analytic model to make inferences about future and past states. (See definition of “analytic model.”)

**Analytic model**—a formal representation of a system or process, including its structure, factors of interest, and assumptions that can be used to infer the system’s future outcomes or infer its past causes. Examples include causal models, simulation models, and statistical models.

**Causal analysis**—an analytic method whose primary focus is the causal relationships among factors of interest within a process or capability (i.e., causal discovery). Causal analysis is often employed in support of understanding how a problem arose (i.e., root cause analysis) or in estimating the impacts that certain changes would have on future outcomes (i.e., causal inference).

**Controls**—the means of managing risk, manual or automated, including policies, procedures, guidelines, practices, or organizational structures, which can be of an administrative, technical, management, or legal nature. Controls help ensure the safety, accuracy, and reliability of information systems; the safeguarding of assets; the promotion of administrative efficiency; and adherence to standards. [Caralli 2016, Ross 2021]

**Explainable AI**—a system that delivers or contains evidence and/or reason(s) for outputs and/or processes which is understandable and accurately reflects the systems’ processes. (Adapted from Phillips et. al. [Phillips 2021].) Stakeholders have different needs for explainability (e.g., end users vs. engineers) and will need different levels of fidelity for the provided evidence.

**Operational experimentation**—testing early-stage proofs of concepts and prototypes by asking the expected end users to attempt to complete intended tasks in their context of use (i.e., in the field). This method provides the development team with early feedback and minimizes risk in development.

**Operational risk**—the potential impact on organizational services and stakeholders resulting from failed or inadequate processes, systems, or technology; or the deliberate or mistaken actions of people; or external events. (Adapted from the CERT-RMM [Caralli 2016].)

**Organizational unit (OU)**—the particular group or component of the organization that is selected to be assessed. In a small-to-medium-sized organization, the OU may be the whole organization, but in a large organization, the OU may be a particular technology, product, or service group of the larger organization.

**Responsible AI (RAI)**—principles and methods that guide teams in the intentional design, development, deployment, and use of AI systems. The objectives of RAI are to minimize and mitigate risk to people, improve the trustworthiness of AI, and maximize the benefits of AI’s capability to augment humans.

**Structured data**—information that is organized according to an explicitly defined data model, schema, or structure (i.e., rows and columns), or a relational database.

**Transparent AI**—a system that discloses relevant information, in context, about the AI portions of the system so that the system can be assessed. Disclosures for transparency need to be relevant to stakeholders (e.g., end users vs. engineers) and may include information about the source(s) of data, algorithms in use, monitoring information, and more.

**Unstructured data**—information that is not organized in a predefined manner, may have diverse structures, and may have multiple formats including text, images, video, and audio files.

**Use case**—a specific goal or task that a user seeks to accomplish defined in terms of the desired outcome rather than the steps taken to achieve it. Use cases and workflows may have a many-to-many relationship.

**Workflow**—an end-to-end business or technical process to address a use case(s).

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